

Title **Modbus Parameter List**
Subject **Modbus Register Definition**
Project **NBEGESAM**
PID **226012**

Abstract Definition of Modbus Objects

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	1.2	20-Aug-2013	Constantin Philipp	Status=Final - valid without Approval
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Modbus Object	RVS61	RVS21	MdB Register (see sheet "ProtocolIDDescription" Address [dec])	R/W	Parameter Number	Menu	Parameter	Enumeration/ Description	Lower Limit	Upper Limit	Unit	Resolution
x	x	x	0x0400...0x0	BL			Heating circuit 1					
x	x	x	1024	R/W	700		Operating Mode	0: Protection x 1: Automatic 2: Reduced 3: Comfort				
x	x	x	1025	R/W	710		Comfort setpoint		4	35	°C	1/64
x	x	x	1026	R/W	712		Reduced setpoint		4	35	°C	1/64
x	x	x	1027	R/W	714		Frost protection setpoint		4	35	°C	1/64
x	x	x	1028	R/W	720		Heating curve slope		0.1	4		1/50
x	x	x	1029	R/W	721		Heating curve displacement		-4.5	4.5	°C	1/64
x	x	x	1030	R/W	730		Summer/winter heating limit		8	30	°C	1/64
x	x	x	1031	R/W	730		Status/Comand	See sheet "Status_Command" Status: (Read) Bit0: ObjectUnused Bit1: SensorShortCircuit Bit6: DeviceFailure Comand: (Write) 0: Release 1: Write 5: SetOutOfService 6: ResetOutOfService				
x	x	x	1032	R/W	732		24-hour heating limit		-10	10	°C	1/64
x	x	x	1033	R/W	732		Status/Comand	see description				
x	x	x	1034	R/W	740		Flow temp setpoint min		8	95	°C	1/64
x	x	x	1035	R/W	741		Flow temp setpoint max		8	95	°C	1/64
x	x	x	1036	R/W	742		Flow temp setpoint room stat		8	95	°C	1/64
x	x	x	1037	R/W	742		Status/Comand	see description				
x	x	x	1038	R/W	750		Room influence		1	100	%	1
x	x	x	1039	R/W	750		Status/Comand	see description				
x	x	x	1040	R	8739		Relative room humidity 1		0	100	%	1
x	x	x	1041	R	8739		Status	Status: (Read) Bit0: ObjectUnused Bit1: SensorShortCircuit Bit6: DeviceFailure				
x	x	x	1042	R	8740		Room temperature 1		0	50	°C	1/64
x	x	x	1043	R	8740		Status	see description				
x	x	x	1044	R	8741		Room setpoint 1		4	35	°C	1/64
x	x	x	1045	R	8741		Status	see description				
x	x	x	1046	R	8743		Flow temperature 1		0	140	°C	1/64
x	x	x	1047	R	8743		Status	see description				
x	x	x	1048	R	8744		Flow temperature setpoint 1		0	140	°C	1/64
x	x	x	1049	R	8744		Status	see description				
x	x	x	1050	R	8749		Room thermostat 1	x 0: No demand 1: Demand	0	1		
x	x	x	1051	R	8749		Status	see description				
x	x	x	1052	R/W	6740		Flow temp 1 alarm		10	240	min	1
x	x	x	1053	R/W	6740		Status/Comand	see description				
x	x	x	1054	R	8000		State heating circuit 1	see description: status codes	0	1000		
x	x	x	0x0800...0x0	BL			Cooling circuit 1					
x	x	x	2048	R/W	901		Operating mode	0: Off x 1: Automatic	0	1		
x	x	x	2049	R/W	902		Comfort setpoint		15	40	°C	1/64
x	x	x	2050	R/W	907		Release	x 1: 24h/day	1	3		
x	x	x	2051	R/W	908		Flow temp setp at OT 25°C		6	35	°C	1/64
x	x	x	2052	R/W	909		Flow temp setp at OT 35°C		6	35	°C	1/64
x	x	x	2053	R/W	912		Cooling limit at OT		8	35	°C	1/64
x	x	x	2054	R/W	912		Status/Comand	see description				
x	x	x	2055	R/W	913		Lock time at end of heating		8	100	h	1
x	x	x	2056	R/W	913		Status/Comand	see description				
x	x	x	2057	R/W	918		Summer comp start at OT		20	50	°C	1/64
x	x	x	2058	R/W	919		Summer comp end at OT		20	50	°C	1/64
x	x	x	2059	R/W	920		Summer comp setp increase		1	10	°C	1/64
x	x	x	2060	R/W	920		Status/Comand	see description				
x	x	x	2061	R/W	923		Flow temp setp min OT 25°C		6	35	°C	1/64
x	x	x	2062	R/W	924		Flow temp setp min OT 35°C		6	35	°C	1/64
x	x	x	2063	R/W	928		Room influence		1	100	%	1
x	x	x	2064	R/W	928		Status/Comand	see description				
x	x	x	2065	R	8747		Dewpoint temp 1		0	50	°C	1/64
x	x	x	2066	R	8747		Status/Comand	see description				
x	x	x	2067	R	8756		Flow temp cooling 1		0	140	°C	1/64
x	x	x	2068	R	8756		Status	see description				
x	x	x	2069	R	8757		Flow temp setp cooling 1		0	140	°C	1/64
x	x	x	2070	R	8757		Status	see description				
x	x	x	2071	R/W	6746		Flow temp cooling 1 alarm		10	240	min	1

x	x	x	2072	R/W	6740	Status/Comand	see description				
x	x	x	2073	R	8004	State cooling circuit 1	see description: status codes	0	1000		
x	x	x	0x1000...0x1	BL		Heating circuit 2					
x	x	x	4096	R/W	1000	Op.mode	0: Protection x 1: Automatic 2: Reduced 3: Comfort				
x	x	x	4097	R/W	1010	Comfort setpoint		4	35	°C	1/64
x	x	x	4098	R/W	1012	Reduced setpoint		4	35	°C	1/64
x	x	x	4099	R/W	1014	Frost protection setpoint		4	35	°C	1/64
x	x	x	4100	R/W	1020	Heating curve slope		0.1	4		1/50
x	x	x	4101	R/W	1021	Heating curve displacement		-4.5	4.5	°C	1/64
x	x	x	4102	R/W	1030	Summer/winter heating limit		8	30	°C	1/64
x	x	x	4103	R/W	1030	Status/Comand	see description				
x	x	x	4104	R/W	1032	24-hour heating limit		-10	10	°C	1/64
x	x	x	4105	R/W	1032	Status/Comand	see description				
x	x	x	4106	R/W	1040	Flow temp setpoint min		8	95	°C	1/64
x	x	x	4107	R/W	1041	Flow temp setpoint max		8	95	°C	1/64
x	x	x	4108	R/W	1042	Flow temp setpoint room stat		8	95	°C	1/64
x	x	x	4109	R/W	1042	Status/Comand	see description				
x	x	x	4110	R/W	1050	Room influence		1	100	%	1
x	x	x	4111	R/W	1050	Status/Comand	see description				
x	x	x	4112	R	8769	Relative room humidity 2		0	100	%	1
x	x	x	4113	R	8769	Status	see description				
x	x	x	4114	R	8770	Room temp 2		0	50	°C	1/64
x	x	x	4115	R	8770	Status	see description				
x	x	x	4116	R	8771	Room setpoint 2		4	35	°C	1/64
x	x	x	4117	R	8771	Status	see description				
x	x	x	4118	R	8773	Flow temp 2		0	140	°C	1/64
x	x	x	4119	R	8773	Status	see description				
x	x	x	4120	R	8774	Flow temp setpoint 2		0	140	°C	1/64
x	x	x	4121	R	8774	Status	see description				
x	x	x	4122	R	8779	Room thermostat 2	x 0: No demand 1: Demand	0	1		
x	x	x	4123	R	8779	Status	see description				
x	x	x	4124	R/W	6741	Flow temp 2 alarm		10	240	min	1
x	x	x	4125	R/W	6741	Status/Comand	see description				
x	x	x	4126	R	8001	State heating circuit 2	see description: status codes	0	1000		
x	x	x	0x1400...0x1	BL		Cooling circuit 2					
x	x	x	5120	R/W	1201	Operating mode cooling circuit 2	0: Off x 1: Automatic	0	1		
x	x	x	5121	R/W	1202	Room temp Comfort setpoint cooling circuit 2		15	40	°C	1/64
x	x	x	5122	R/W	1207	Release	x 1: 24h/day	1	3		
x	x	x	5123	R/W	1208	Flow setpoint at outside temp 25°C CC2		6	35	°C	1/64
x	x	x	5124	R/W	1209	Flow setpoint at outside temp 35°C CC2		6	35	°C	1/64
x	x	x	5125	R/W	1212	Cooling limit at outs temp cooling circuit 2		8	35	°C	1/64
x	x	x	5126	R/W	1212	Status/Comand	see description				
x	x	x	5127	R/W	1213	Remaining time after heating cooling circuit 2		8	100	h	1
x	x	x	5128	R/W	1213	Status/Comand	see description				
x	x	x	5129	R/W	1218	Summer comp start at OT		20	50	°C	1/64
x	x	x	5130	R/W	1219	Summer comp end at OT		20	50	°C	1/64
x	x	x	5131	R/W	1220	Summer comp setp increase		1	10	°C	1/64
x	x	x	5132	R/W	1220	Status/Comand	see description				
x	x	x	5133	R/W	1223	Flow setpoint min at outside temp 25°C CC2		6	35	°C	1/64
x	x	x	5134	R/W	1224	Flow setpoint min at outside temp 35°C CC2		6	35	°C	1/64
x	x	x	5135	R/W	1228	Room temperature gain factor cooling circuit 2		1	100	%	1
x	x	x	5136	R/W	1228	Status/Comand	see description				
x	x	x	5137	R	8777	Dewpoint temp 2		0	50	°C	1/64
x	x	x	5138	R	8777	Status/Comand	see description				
x	x	x	5139	R	8786	Flow temp cooling 2		0	140	°C	1/64
x	x	x	5140	R	8786	Status	see description				
x	x	x	5141	R	8787	Flow temp setp cooling 2		0	140	°C	1/64
x	x	x	5142	R	8787	Status	see description				
x	x	x	5143	R/W	6747	Flow temp cooling 2 alarm		10	240	min	1
x	x	x	5144	R/W	6747	Status/Comand	see description				
x	x	x	5145	R	8025	State cooling circuit 2	see description: status codes	0	1000		
x	x	x	0x1C00...0x1	BL		Heating circuit 3					
x	x	x	7168	R/W	1300	Operating mode	0: Protection x 1: Automatic 2: Reduced 3: Comfort	0	3		
x	x	x	7169	R/W	1310	Comfort setpoint		4	35	°C	1/64
x	x	x	7170	R/W	1312	Reduced setpoint		4	35	°C	1/64
x	x	x	7171	R/W	1314	Frost protection setpoint		4	35	°C	1/64
x	x	x	7172	R/W	1320	Heating curve slope		0.1	4		1/50
x	x	x	7173	R/W	1321	Heating curve displacement		-4.5	4.5	°C	1/64
x	x	x	7174	R/W	1330	Summer/winter heating limit		8	30	°C	1/64
x	x	x	7175	R/W	1330	Status/Comand	see description				
x	x	x	7176	R/W	1332	24-hour heating limit		-10	10	°C	1/64
x	x	x	7177	R/W	1332	Status/Comand	see description				
x	x	x	7178	R/W	1340	Flow temp setpoint min		8	95	°C	1/64
x	x	x	7179	R/W	1341	Flow temp setpoint max		8	95	°C	1/64
x	x	x	7180	R/W	1342	Flow temp setpoint room stat		8	95	°C	1/64
x	x	x	7181	R/W	1342	Status/Comand	see description				
x	x	x	7182	R/W	1350	Room influence		1	100	%	1
x	x	x	7183	R/W	1350	Status/Comand	see description				
x	x	x	7184	R	8799	Relative room humidity 3		0	100	%	1

x	x	x	7185	R	8799	Status	see description				
x	x	x	7186	R	8800	Room temp 3		0	50	°C	1/64
x	x	x	7187	R	8800	Status	see description				
x	x	x	7188	R	8801	Room setpoint 3		4	35	°C	1/64
x	x	x	7189	R	8801	Status	see description				
x	x	x	7190	R	8803	Flow temp setpoint 3		0	140	°C	1/64
x	x	x	7191	R	8803	Status	see description				
x	x	x	7192	R	8804	Flow temp 3		0	140	°C	1/64
x	x	x	7193	R	8804	Status	see description				
x	x	x	7194	R	8809	Room thermostat 3	x 0: No demand 1: Demand	0	1		
x	x	x	7195	R	8809	Status	see description				
x	x	x	7196	R/W	6742	Flow temp 3 alarm		10	240	min	1
x	x	x	7197	R/W	6742	Status/Comand	see description				
x	x	x	7198	R	8002	State heating circuit 3	see description: status codes	0	1000		
x	x	x	0x2000...0x2	BL		Cooling circuit 3					
x	x	x	8192	R/W	1501	Operating mode cooling circuit 3	0: Off x 1: Automatic	0	1		
x	x	x	8193	R/W	1502	Room temp Comfort setpoint cooling circuit 3		17	40	°C	1/64
x	x	x	8194	R/W	1507	Release	x 1: 24h/day	1	3		
x	x	x	8195	R/W	1508	Flow setpoint at outside temp 35°C CC3		6	35	°C	1/64
x	x	x	8196	R/W	1509	Flow setpoint at outside temp 35°C CC3		6	35	°C	1/64
x	x	x	8197	R/W	1512	Cooling limit at outs temp cooling circuit 3		8	35	°C	1/64
x	x	x	8198	R/W	1512	Status/Comand	see description				
x	x	x	8199	R/W	1513	Remaining time after heating cooling circuit 3		8	100	h	1
x	x	x	8200	R/W	1513	Status/Comand	see description				
x	x	x	8201	R/W	1518	Summer comp start at OT		20	50	°C	1/64
x	x	x	8202	R/W	1519	Summer comp end at OT		20	50	°C	1/64
x	x	x	8203	R/W	1520	Summer comp setp increase		1	10	°C	1/64
x	x	x	8204	R/W	1520	Status/Comand	see description				
x	x	x	8205	R/W	1523	Flow setpoint min at outside temp 35°C CC3		6	35	°C	1/64
x	x	x	8206	R/W	1524	Flow setpoint min at outside temp 35°C CC3		6	35	°C	1/64
x	x	x	8207	R/W	1528	Room temperature gain factor cooling circuit 3		1	100	%	1
x	x	x	8208	R/W	1528	Status/Comand	see description				
x	x	x	8209	R	8807	Dewpoint temp 3		0	50	°C	1/64
x	x	x	8210	R	8807	Status/Comand	see description				
x	x	x	8211	R	8816	Flow temp cooling 3		0	140	°C	1/64
x	x	x	8212	R	8816	Status	see description				
x	x	x	8213	R	8817	Flow temp setp cooling 3		0	140	°C	1/64
x	x	x	8214	R	8817	Status	see description				
x	x	x	8215	R/W	6748	Flow temp cooling 3 alarm		10	240	min	1
x	x	x	8216	R/W	6748	Status/Comand	see description				
x	x	x	8217	R	8026	State cooling circuit 3	see description: status codes	0	1000		
x	x	x	0x2800...0x2	BL		Domestic hot water					
x	x	x	10240	R/W	1600	DHW Operating mode	0: Off x 1: On 2: Eco	0	2		
x	x	x	10241	R/W	1610	Nominal setpoint		8	80	°C	1/64
x	x	x	10242	R/W	1612	Reduced setpoint		8	80	°C	1/64
x	x	x	10243	R/W	1620	Release	0: 24h/day 1: Time programs HCs x 2: Time program 4/DHW 3: Low-tariff 4: Tprog 4/DHW or low-tariff	0	4		
x	x	x	10244	R/W	1640	Legionella function	x 0: Off 1: Periodically 2: Fixed weekday	0	2		
x	x	x	10245	R/W	1641	Legionella funct periodically		1	7		
x	x	x	10246	R/W	1642	Legionella funct weekday	x 1: Monday 2: Tuesday 3: Wednesday 4: Thursday 5: Friday 6: Saturday 7: Sunday	1	7		
x	x	x	10247	R/W	1644	Legionella funct time		00:00	23:50	h:m	10min
x	x	x	10248	R/W	1644	Status/Comand	see description				
x	x	x	10249	R/W	1645	Legionella funct setpoint		55	95	°C	1/64
x	x	x	10250	R/W	1646	Legionella funct duration		2	360	min	1
x	x	x	10251	R/W	1646	Status/Comand	see description				
x	x	x	10252	R/W	6745	DHW charging alarm		1	48	h	1
x	x	x	10253	R/W	6745	Status/Comand	see description				
x	x	x	10254	R	8003	State DHW	see description: status codes	0	1000		
x	x	x	0x2C00...0x2	BL		DHW storage tank					
x	x	x	11264	R	8830	DHW temp 1		0	140	°C	1/64
x	x	x	11265	R	8830	Status	see description				
x	x	x	11266	R	8832	DHW temp 2		0	140	°C	1/64
x	x	x	11267	R	8832	Status	see description				
x	x	x	11268	R/W	8840	Hours run DHW pump		0	199'999	h	1/3600
			11269			(32 Bit Data)					
x	x	x	11270	R/W	8841	Start counter DHW pump		0	199999		1
			11271			(32 Bit Data)					
x	x	x	11272	R/W	8842	Hours run el DHW		0	199'999	h	1/3600
			11273			(32 Bit Data)					
x	x	x	11274	R/W	8843	Start counter el DHW		0	199999		1
			11275			(32 Bit Data)					
x	x	x	11276	R/W	7091	DHW stor tank since maint		0	240	Months	1

x	x	x	11277	R/W	7092	DHW chrg temp HP min		8	80	°C	1/64
x	x	x	11278	R/W	7092	Status/Comand	see description				
x	x	x	11279	R/W	7093	Curr DHW chrg temp HP		8	80	°C	1/64
x	x	x	0x3400...0x3	BL		Consumer circuit 1					
x	x	x	13312	R/W	1854	Request opt energy	x 0: Off 1: On	0	1		
x	x	x	13313	R/W	1859	Flow temp setp cons request		8	120	°C	1/64
x	x	x	0x3800...0x3	BL		Consumer circuit 2					
x	x	x	14336	R/W	1904	Request opt energy	x 0: Off 1: On	0	1		
x	x	x	14337	R/W	1909	Flow temp setp cons request		8	120	°C	1/64
x	x	x	0x3C00...0x3	BL		Swimming pool					
x	x	x	15360	R/W	1959	Flow temp setpoint		8	120	°C	1/64
x	x	x	15361	R/W	2055	Setpoint solar heating		8	80	°C	1/64
x	x	x	15362	R/W	2056	Setpoint source heating		8	35	°C	1/64
x	x	x	15363	R	8900	Swimming pool temp		0	140	°C	1/64
x	x	x	15364	R	8900	Status	see description				
x	x	x	15365	R	8901	Swimming pool setpoint		0	80	°C	1/64
x	x	x	15366	R	8901	Status	see description				
x	x	x	15367	R	8011	State swimming pool	see description: status codes	0	1000		
x	x	x	0x4400...0x4	BL		Buffer storage tank					
x	x	x	17408	R	8981	Buffer setpoint		0	140	°C	1/64
x	x	x	17409	R	8981	Status/Comand	see description				
x	x	x	17410	R	8980	Buffer temp 1		0	140	°C	1/64
x	x	x	17411	R	8980	Status	see description				
x	x	x	17412	R	8982	Buffer temp 2		0	140	°C	1/64
x	x	x	17413	R	8982	Status	see description				
x	x	x	17414	R/W	8990	Hours run el buffer		0	199'999	h	1/3600
			17415				(32 Bit Data)				
x	x	x	17416	R/W	8991	Start counter el buffer		0	199999		1
			17417				(32 Bit Data)				
x	x	x	17418	R	8010	State buffer	see description: status codes	0	1000		
x	x	x	0x4800...0x4	BL		Common flow					
x	x	x	18432	R	8951	Common flow temp setpoint		0	140	°C	1/64
x	x	x	18433	R	8951	Status	see description				
x	x	x	18434	R	8957	Common flow setp refrig		0	140	°C	1/64
x	x	x	18435	R	8957	Status	see description				
x	x	x	0x5000...0x5	BL		Heat pump					
x	x	x	20480	R	8410	Return temp HP		0	140	°C	1/64
x	x	x	20481	R	8410	Status	see description				
x	x	x	20482	R	8411	Setpoint HP		0	140	°C	1/64
x	x	x	20483	R	8411	Status	see description				
x	x	x	20484	R	8412	Flow temp HP		0	140	°C	1/64
x	x	x	20485	R	8412	Status	see description				
x	x	x	20486	R	8435	Evaporation temp		-50	180	°C	1/64
x	x	x	20487	R	8435	Status	see description				
x	x	x	20488	R	8436	Superheat		-10	180	°C	1/64
x	x	x	20489	R	8436	Status	see description				
x	x	x	20490	R	8436	Superheat setpoint		0	140	°C	1/64
x	x	x	20491	R	8436	Status	see description				
x	x	x	20492	R	8440	Remain sDaye 1 off time min		0	255	min	1
x	x	x	20493	R	8440	Status	see description				
x	x	x	20494	R	8441	Remain sDaye 2 off time min		0	255	min	1
x	x	x	20495	R	8441	Status	see description				
x	x	x	20496	R	8442	Remain sDaye 1 on time min		0	255	min	1
x	x	x	20497	R	8442	Status	see description				
x	x	x	20498	R	8443	Remain sDaye 2 on time min		0	255	min	1
x	x	x	20499	R	8443	Status	see description				
x	x	x	20500	R	8444	Remain limit source temp		0	65'535	h:m	1min
x	x	x	20501	R	8444	Status	see description				
x	x	x	20502	R	8446	Compressor sequence	x 0: 1 - 2 1: 2 - 1	0	1		
x	x	x	20503	R/W	8449	Hours run refrigerant circuit		0	199'999	h	1/3600
			20504				(32 Bit Data)				
x	x	x	20505	R/W	8450	Hours run compressor 1		0	199'999	h	1/3600
			20506				(32 Bit Data)				
x	x	x	20507	R/W	8451	Start counter compressor 1		0	199999		1
			20508				(32 Bit Data)				
x	x		20509	R/W	8452	Hours run compressor 2		0	199'999	h	1/3600
			20510				(32 Bit Data)				
x	x		20511	R/W	8453	Start counter compressor 2		0	199999		1
			20512				(32 Bit Data)				
x	x	x	20513	R/W	8454	Locking time HP		0	199'999	h	1/3600
			20514				(32 Bit Data)				
x	x	x	20515	R/W	8455	Counter number of locks HP		0	199999		1
			20516				(32 Bit Data)				
x	x	x	20517	R/W	8456	Hours run el flow		0	199'999	h	1/3600
			20518				(32 Bit Data)				
x	x	x	20519	R/W	8457	Start counter el flow		0	199999		1
			20520				(32 Bit Data)				
x	x	x	20521	R/W	8458	Hours run el 2 flow		0	199'999	h	1/3600
			20522				(32 Bit Data)				
x	x	x	20523	R/W	8459	Start counter el 2 flow		0	199999		1
			20524				(32 Bit Data)				
x	x	x	20525	R	8463	Evaporation temp EVI		-50	180	°C	1/64
x	x	x	20526	R	8463	Status	see description				
x	x	x	20527	R	8464	Superheat EVI		-10	180	°C	1/64

x	x	x	20528	R	8464	Status	see description				
x	x	x	20529	R	8464	Superheat setpoint EVI		0	140	°C	1/64
x	x	x	20530	R	8464	Status	see description				
x	x	x	20531	R	8477	Temp diff defrost act value		-50	50	°C	1/64
x	x	x	20532	R	8477	Status	see description				
x	x	x	20533	R	8478	Temp diff defrost setpoint		-50	50	°C	1/64
x	x	x	20534	R	8478	Status	see description				
x	x	x	20535	R	8480	Remain time defrost lock		0	255	min	1
x	x	x	20536	R	8480	Status	see description				
x	x	x	20537	R	8481	Remain time forced defrost		0	65'535	h:m	1min
x	x	x	20538	R	8481	Status	see description				
x	x	x	20539	R	8482	Remain time defrost settling		0	255	min	1
x	x	x	20540	R	8482	Status	see description				
x	x	x	20541	R	8485	Number defrost attempts		0	10		
x	x	x	20542	R	8487	Defrost state	x 0: HP off, defr release OT off 1: Locked 2: Monitoring ice 3: Preheating for defrost 4: Defrost active 5: Dripping 6: Cooling down evaporator 7: Fault 8: Forced defrost 9: Defrost settling 10: Defrost with fan 11: Defrost with compressor 12: Forced defrost fan 13: Forced defrost compressor 14: Start delay defrost	0	14		
x	x	x	20543	R/W	2909	Release below outside temp		-50	50	°C	1/64
x	x	x	20544	R/W	2909	Status/Comand	see description				
x	x	x	20545	R/W	2910	Release above outside temp		-30	30	°C	1/64
x	x	x	20546	R/W	2910	Status/Comand	see description				
x	x	x	20547	R/W	6711	Reset HP	x 0: No 1: Yes	0	1		
x	x	x	20548	R/W	7071	HP time since maint		0	240	Months	1
x	x	x	20549	R/W	7073	Cur starts compr1/hrs run		0	12		1/10
x	x	x	20550	R/W	7075	Cur starts compr2/hrs run		0	12		1/10
x	x	x	20551	R/W	7077	Cur diff condens max/week		0	250		
x	x	x	20552	R/W	7079	Cur diff condens min/week		0	250		
x	x	x	20553	R/W	7081	Cur diff evap max/week		0	250		
x	x	x	20554	R/W	7083	Cur diff evap min/week		0	250		
x	x	x	20555	R/W	7152	Triggering defrost	x 0: No	0	1		
x	x	x	20556	R	8006	State heat pump	see description: status codes	0	1000		
x	x	x	0x7400...0x7	BL		Energy meter					
x	x	x	29696	R	3110	Heat delivered		0	9999999	kWh	1
			29697			(32 Bit Data)					
x	x	x	29698	R	3110	Status/Comand	see description				
x	x	x	29699	R	3113	Energy brought in		0	3500000	kWh	1
			29700			(32 Bit Data)					
x	x	x	29701	R	3113	Status/Comand	see description				
x	x	x	29702	R	3116	Performance factor		0	10		1/100
x	x	x	29703	R	3116	Status	see description				
x	x	x	29704	R	3120	Yearly perf factor 1		0	10		1/100
x	x	x	29705	R	3120	Status	see description				
x	x	x	29706	R	3120	Fixed day 1	Year	100	199	Year	1
x	x	x	29707	R	3120		Month	1	12	Month	1
x	x	x	29708	R	3120		Day	1	31	Day	1
x	x	x	29709	R	3120	Status	see description				
x	x	x	29710	R	3121	Heat delivered heating 1		0	9999999	kWh	1
			29711			(32 Bit Data)					
x	x	x	29712	R	3121	Status	see description				
x	x	x	29713	R	3122	Heat delivered DHW 1		0	9999999	kWh	1
			29714			(32 Bit Data)					
x	x	x	29715	R	3122	Status	see description				
x	x	x	29716	R	3124	Energy brought in heating 1		0	3500000	kWh	1
			29717			(32 Bit Data)					
x	x	x	29718	R	3124	Status	see description				
x	x	x	29719	R	3125	Energy brought in DHW 1		0	3500000	kWh	1
			29720			(32 Bit Data)					
x	x	x	29721	R	3125	Status	see description				
x	x	x	29722	R	3127	Yearly perf factor 2		0	10		1/100
x	x	x	29723	R	3127	Status	see description				
x	x	x	29724	R	3127	Fixed day 2	Year	100	199	Year	1
x	x	x	29725	R	3127		Month	1	12	Month	1
x	x	x	29726	R	3127		Day	1	31	Day	1
x	x	x	29727	R	3127	Status	see description				
x	x	x	29728	R	3128	Heat delivered heating 2		0	9999999	kWh	1
			29729			(32 Bit Data)					
x	x	x	29730	R	3128	Status	see description				
x	x	x	29731	R	3129	Heat delivered DHW 2		0	9999999	kWh	1
			29732			(32 Bit Data)					
x	x	x	29733	R	3129	Status	see description				
x	x	x	29734	R	3131	Energy brought in heating 2		0	3500000	kWh	1
			29735			(32 Bit Data)					
x	x	x	29736	R	3131	Status	see description				
x	x	x	29737	R	3132	Energy brought in DHW 2		0	3500000	kWh	1

			29738				(32 Bit Data)					
x	x	x	29739	R	3132	Status	see description					
x	x	x	29740	R	3134	Yearly perf factor 3		0	10			1/100
x	x	x	29741	R	3134	Status	see description					
x	x	x	29742	R	3134	Fixed day 3	Year	100	199	Year		1
x	x	x	29743	R	3134		Month	1	12	Month		1
x	x	x	29744	R	3134		Day	1	31	Day		1
x	x	x	29745	R	3134	Status	see description					
x	x	x	29746	R	3135	Heat delivered heating 3		0	9999999	kWh		1
			29747				(32 Bit Data)					
x	x	x	29748	R	3135	Status	see description					
x	x	x	29749	R	3136	Heat delivered DHW 3		0	9999999	kWh		1
			29750									
x	x	x	29751	R	3136	Status	see description					
x	x	x	29752	R	3138	Energy brought in heating 3		0	3500000	kWh		1
			29753				(32 Bit Data)					
x	x	x	29754	R	3138	Status	see description					
x	x	x	29755	R	3139	Energy brought in DHW 3		0	3500000	kWh		1
			29756				(32 Bit Data)					
x	x	x	29757	R	3139	Status	see description					
x	x	x	29758	R	3141	Yearly perf factor 4		0	10			1/100
x	x	x	29759	R	3141	Status	see description					
x	x	x	29760	R	3141	Fixed day 4	Year	100	199	Year		1
x	x	x	29761	R	3141		Month	1	12	Month		1
x	x	x	29762	R	3141		Day	1	31	Day		1
x	x	x	29763	R	3141	Status	see description					
x	x	x	29764	R	3142	Heat delivered heating 4		0	9999999	kWh		1
			29765				(32 Bit Data)					
x	x	x	29766	R	3142	Status	see description					
x	x	x	29767	R	3143	Heat delivered DHW 4		0	9999999	kWh		1
			29768				(32 Bit Data)					
x	x	x	29769	R	3143	Status	see description					
x	x	x	29770	R	3145	Energy brought in heating 4		0	3500000	kWh		1
			29771				(32 Bit Data)					
x	x	x	29772	R	3145	Status	see description					
x	x	x	29773	R	3146	Energy brought in DHW 4		0	3500000	kWh		1
			29774				(32 Bit Data)					
x	x	x	29775	R	3146	Status	see description					
x	x	x	29776	R	3148	Yearly perf factor 5		0	10			1/100
x	x	x	29777	R	3148	Status	see description					
x	x	x	29778	R	3148	Fixed day 5	Year	100	199	Year		1
x	x	x	29779	R	3148		Month	1	12	Month		1
x	x	x	29780	R	3148		Day	1	31	Day		1
x	x	x	29781	R	3148	Status	see description					
x	x	x	29782	R	3149	Heat delivered heating 5		0	9999999	kWh		1
			29783				(32 Bit Data)					
x	x	x	29784	R	3149	Status	see description					
x	x	x	29785	R	3150	Heat delivered DHW 5		0	9999999	kWh		1
			29786				(32 Bit Data)					
x	x	x	29787	R	3150	Status	see description					
x	x	x	29788	R	3152	Energy brought in heating 5		0	3500000	kWh		1
			29789				(32 Bit Data)					
x	x	x	29790	R	3152	Status	see description					
x	x	x	29791	R	3153	Energy brought in DHW 5		0	3500000	kWh		1
			29792				(32 Bit Data)					
x	x	x	29793	R	3153	Status	see description					
x	x	x	29794	R	3155	Yearly perf factor 6		0	10			1/100
x	x	x	29795	R	3155	Status	see description					
x	x	x	29796	R	3155	Fixed day 6	Year	100	199	Year		1
x	x	x	29797	R	3155		Month	1	12	Month		1
x	x	x	29798	R	3155		Day	1	31	Day		1
x	x	x	29799	R	3155	Status	see description					
x	x	x	29800	R	3156	Heat delivered heating 6		0	9999999	kWh		1
			29801				(32 Bit Data)					
x	x	x	29802	R	3156	Status	see description					
x	x	x	29803	R	3157	Heat delivered DHW 6		0	9999999	kWh		1
			29804				(32 Bit Data)					
x	x	x	29805	R	3157	Status	see description					
x	x	x	29806	R	3159	Energy brought in heating 6		0	3500000	kWh		1
			29807				(32 Bit Data)					
x	x	x	29808	R	3159	Status	see description					
x	x	x	29809	R	3160	Energy brought in DHW 6		0	3500000	kWh		1
			29810				(32 Bit Data)					
x	x	x	29811	R	3160	Status	see description					
x	x	x	29812	R	3162	Yearly perf factor 7		0	10			1/100
x	x	x	29813	R	3162	Status	see description					
x	x	x	29814	R	3162	Fixed day 7	Year	100	199	Year		1
x	x	x	29815	R	3162		Month	1	12	Month		1
x	x	x	29816	R	3162		Day	1	31	Day		1
x	x	x	29817	R	3162	Status	see description					
x	x	x	29818	R	3163	Heat delivered heating 7		0	9999999	kWh		1
			29819				(32 Bit Data)					
x	x	x	29820	R	3162	Status	see description					
x	x	x	29821	R	3164	Heat delivered DHW 7		0	9999999	kWh		1
			29822				(32 Bit Data)					
x	x	x	29823	R	3164	Status	see description					
x	x	x	29824	R	3166	Energy brought in heating 7		0	3500000	kWh		1

			29825				(32 Bit Data)					
x	x	x	29826	R	3166	Status	see description					
x	x	x	29827	R	3167	Energy brought in DHW 7		0	3500000	kWh	1	
			29828				(32 Bit Data)					
x	x	x	29829	R	3167	Status	see description					
x	x	x	29830	R	3169	Yearly perf factor 8		0	10		1/100	
x	x	x	29831	R	3169	Status	see description					
x	x	x	29832	R	3169	Fixed day 8	Year	100	199	Year	1	
x	x	x	29833	R	3169		Month	1	12	Month	1	
x	x	x	29834	R	3169		Day	1	31	Day	1	
x	x	x	29835	R	3169	Status	see description					
x	x	x	29836	R	3170	Heat delivered heating 8		0	9999999	kWh	1	
			29837				(32 Bit Data)					
x	x	x	29838	R	3170	Status	see description					
x	x	x	29839	R	3171	Heat delivered DHW 8		0	9999999	kWh	1	
			29840				(32 Bit Data)					
x	x	x	29841	R	3171	Status	see description					
x	x	x	29842	R	3173	Energy brought in heating 8		0	3500000	kWh	1	
			29843				(32 Bit Data)					
x	x	x	29844	R	3173	Status	see description					
x	x	x	29845	R	3174	Energy brought in DHW 8		0	3500000	kWh	1	
			29846				(32 Bit Data)					
x	x	x	29847	R	3174	Status	see description					
x	x	x	29848	R	3176	Yearly perf factor 9		0	10		1/100	
x	x	x	29849	R	3176	Status	see description					
x	x	x	29850	R	3176	Fixed day 9	Year	100	199	Year	1	
x	x	x	29851	R	3176		Month	1	12	Month	1	
x	x	x	29852	R	3176		Day	1	31	Day	1	
x	x	x	29853	R	3176	Status	see description					
x	x	x	29854	R	3177	Heat delivered heating 9		0	9999999	kWh	1	
			29855				(32 Bit Data)					
x	x	x	29856	R	3177	Status	see description					
x	x	x	29857	R	3178	Heat delivered DHW 9		0	9999999	kWh	1	
			29858				(32 Bit Data)					
x	x	x	29859	R	3178	Status	see description					
x	x	x	29860	R	3180	Energy brought in heating 9		0	3500000	kWh	1	
			29861				(32 Bit Data)					
x	x	x	29862	R	3180	Status	see description					
x	x	x	29863	R	3181	Energy brought in DHW 9		0	3500000	kWh	1	
			29864				(32 Bit Data)					
x	x	x	29865	R	3181	Status	see description					
x	x	x	29866	R	3183	Yearly perf factor 10		0	10		1/100	
x	x	x	29867	R	3183	Status	see description					
x	x	x	29868	R	3183	Fixed day 10	Year	100	199	Year	1	
x	x	x	29869	R	3183		Month	1	12	Month	1	
x	x	x	29870	R	3183		Day	1	31	Day	1	
x	x	x	29871	R	3183	Status	see description					
x	x	x	29872	R	3184	Heat delivered heating 10		0	9999999	kWh	1	
			29873				(32 Bit Data)					
x	x	x	29874	R	3184	Status	see description					
x	x	x	29875	R	3185	Heat delivered DHW 10		0	9999999	kWh	1	
			29876				(32 Bit Data)					
x	x	x	29877	R	3185	Status	see description					
x	x	x	29878	R	3187	Energy brought in heat 10		0	3500000	kWh	1	
			29879				(32 Bit Data)					
x	x	x	29880	R	3187	Status	see description					
x	x	x	29881	R	3188	Energy brought in DHW 10		0	3500000	kWh	1	
			29882				(32 Bit Data)					
x	x	x	29883	R	3188	Status	see description					
x	x	x	0x7800...0x7	BL		Supplementary source						
x	x	x	30720	R/W	3700	Release below outside temp		-50	50	°C	1/64	
x	x	x	30721	R/W	3700	Status/Comand	see description					
x	x	x	30722	R/W	3701	Release above outside temp		-50	50	°C	1/64	
x	x	x	30723	R/W	3701	Status/Comand	see description					
x	x	x	30724	R	8586	Suppl source setpoint		0	140	°C	1/64	
x	x	x	30725	R	8586	Status/Comand	see description					
x	x	x	30726	R	8022	State supplementary source	see description: status codes	0	1000			
x	x	x	0x7C00...0x7	BL		Solid fuel boiler						
x	x	x	31744	R/W	8570	Hours run solid fuel boiler		0	199'999	h	1/3600	
			31745				(32 Bit Data)					
x	x	x	31746	R	8560	Solid fuel boiler temp		0	140	°C	1/64	
x	x	x	31747	R	8560	Status	see description					
x	x	x	31748	R	8561	Solid fuel boiler setpoint		0	140	°C	1/64	
x	x	x	31749	R	8561	Status	see description					
x	x	x	31750	R	8563	Solid fuel boiler return temp		0	140	°C	1/64	
x	x	x	31751	R	8563	Status	see description					
x	x	x	31752	R	8008	State solid fuel boiler	see description: status codes	0	1000			
x	x	x	0x8000...0x8	BL		Solar						
x	x	x	32768	R/W	3810	Temp diff on		0	40	°C	1/64	
x	x	x	32769	R/W	3811	Temp diff off		0	40	°C	1/64	
x	x	x	32770	R/W	3812	Charg temp min DHW st tank		8	95	°C	1/64	
x	x	x	32771	R/w	3812	Status/Comand	see description					
x	x	x	32772	R/W	3813	Temp diff on buffer		0	40	°C	1/64	
x	x	x	32773	R/W	3813	Status/Comand	see description					
x	x	x	32774	R/W	3814	Temp diff off buffer		0	40	°C	1/64	
x	x	x	32775	R/W	3814	Status/Comand	see description					
x	x	x	32776	R/W	3815	Charging temp min buffer		8	95	°C	1/64	

x	x	x	32777	R/W	3815	Status/Comand	see description				
x	x	x	32778	R/W	3816	Temp diff on swi pool		0	40	°C	1/64
x	x	x	32779	R/W	3816	Status/Comand	see description				
x	x	x	32780	R/W	3817	Temp diff off swi pool		0	40	°C	1/64
x	x	x	32781	R/W	3817	Status/Comand	see description				
x	x	x	32782	R/W	3818	Charging temp min swi pool		8	95	°C	1/64
x	x	x	32783	R/W	3818	Status/Comand	see description				
x	x	x	32784	R	8510	Collector temp 1		-28	350	°C	1/64
x	x	x	32785	R	8510	Status	see description				
x	x	x	32786	R	8519	Solar flow temp		-28	350	°C	1/64
x	x	x	32787	R	8519	Status	see description				
x	x	x	32788	R	8520	Solar return temp		-28	350	°C	1/64
x	x	x	32789	R	8520	Status	see description				
x	x	x	32790	R	8526	24-hour yield solar energy		0	999.9	kWh	1/10
x	x	x	32791	R	8526	Status	see description				
x	x	x	32792	R	8527	Total yield solar energy		0	199999	kWh	1/10
			32793				(32 Bit Data)				
x	x	x	32794	R	8527	Status/Comand	see description				
x	x	x	32795	R/W	8530	Hours run solar yield		0	199'999	h	1/3600
			32796				(32 Bit Data)				
x	x	x	32797	R/W	8531	Hours run collect overtemp		0	199'999	h	1/3600
			32798				(32 Bit Data)				
x	x	x	32799	R	8547	Collector temp 2		-28	350	°C	1/64
x	x	x	32800	R	8547	Status	see description				
x	x	x	32801	R	8007	State solar	see description: status codes	0	1000		
x	x	x	0x8900..0x8BFF			Cascade					
x	x	x	34816	R	8100	Priority source 1		0	16		
x	x	x	34817	R	8101	State source 1	x 0: Missing 1: Faulty 2: Manual control active 3: Heat generation lock active 4: Chimney sweep funct active 5: Temporarily unavailable 6: Outside temp limit active 7: Not released 8: Released	0	8		
x	x	x	34818	R	8102	Priority source 2		0	16		
x	x	x	34819	R	8103	State source 2	x 0: Missing 8: Released	0	8		
x	x	x	34820	R	8104	Priority source 3		0	16		
x	x	x	34821	R	8105	State source 3	x 0: Missing 8: Released	0	8		
x	x	x	34822	R	8106	Priority source 4		0	16		
x	x	x	34823	R	8107	State source 4	x 0: Missing 8: Released	0	8		
x	x	x	34824	R	8108	Priority source 5		0	16		
x	x	x	34825	R	8109	State source 5	x 0: Missing 8: Released	0	8		
x	x	x	34826	R	8110	Priority source 6		0	16		
x	x	x	34827	R	8111	State source 6	x 0: Missing 8: Released	0	8		
x	x	x	34828	R	8112	Priority source 7		0	16		
x	x	x	34829	R	8113	State source 7	x 0: Missing 8: Released	0	8		
x	x	x	34830	R	8114	Priority source 8		0	16		
x	x	x	34831	R	8115	State source 8	x 0: Missing 8: Released	0	8		
x	x	x	34832	R	8116	Priority source 9		0	16		
x	x	x	34833	R	8117	State source 9	x 0: Missing 8: Released	0	8		
x	x	x	34834	R	8118	Priority source 10		0	16		
x	x	x	34835	R	8119	State source 10	x 0: Missing 8: Released	0	8		
x	x	x	34836	R	8120	Priority source 11		0	16		
x	x	x	34837	R	8121	State source 11	x 0: Missing 8: Released	0	8		
x	x	x	34838	R	8122	Priority source 12		0	16		
x	x	x	34839	R	8123	State source 12	x 0: Missing 8: Released	0	8		
x	x	x	34840	R	8124	Priority source 13		0	16		
x	x	x	34841	R	8125	State source 13	x 0: Missing 8: Released	0	8		
x	x	x	34842	R	8126	Priority source 14		0	16		

x	x	x	34843	R	8127	State source 14	x 0: Missing 8: Released	0	8		
x	x	x	34844	R	8128	Priority source 15		0	16		
x	x	x	34845	R	8129	State source 15	x 0: Missing 8: Released	0	8		
x	x	x	34846	R	8130	Priority source 16		0	16		
x	x	x	34847	R	8131	State source 16	x 0: Missing 8: Released	0	8		
x	x	x	34848	R	8138	Cascade flow temp		0	140	°C	1/64
x	x	x	34849	R	8138	Status	see description				
x	x	x	34850	R	8139	Cascade flow temp setp		0	140	°C	1/64
x	x	x	34851	R	8139	Status	see description				
x	x	x	34852	R	8150	Source seq ch/over current		0	990	h	1
x	x	x	34853	R	8150	Status	see description				
x	x	x	0x8C00..0x8	BL		General functions					
x	x	x	35840	R/W	5570	Temp diff on dT contr 1		0	40	°C	1/64
x	x	x	35841	R/W	5571	Temp diff off dT contr 1		0	40	°C	1/64
x	x	x	35842	R/W	5572	On temp min dT contr 1		-30	120	°C	1/64
x	x	x	35843	R/W	5580	Temp diff on dT contr 2		0	40	°C	1/64
x	x	x	35844	R/W	5581	Temp diff off dT contr 2		0	40	°C	1/64
x	x	x	35845	R/W	5582	On temp min dT contr 2		-30	120	°C	1/64
x	x	x	35846	R/W	6110	Time constant building		0	50	h	1
x	x	x	35847	R	8565	Flue gas temp		0	350	°C	1/64
x	x	x	35848	R	8565	Status	see description				
x	x	x	35849	R	8567	Flue gas temp max		0	350	°C	1/64
x	x	x	35850	R	8567	Status	see description				
x	x	x	35851	R	8700	Outside temp		-50	50	°C	1/64
x	x	x	35852	R	8700	Status	see description				
x	x	x	35853	R	8701	Outside temp min		-50	50	°C	1/64
x	x	x	35854	R	8702	Outside temp max		-50	50	°C	1/64
x	x	x	35855	R/W	8703	Outside temp attenuated		-50	50	°C	1/64
x	x	x	35856	R	8704	Outside temp composite		-50	50	°C	1/64
x	x	x	35857	R/W	7150	Simulation outside temp		-50	50	°C	1/64
x	x	x	35858	R/W	7150	Status/Comand	see description				
x	x	x	35859	R/W	5600	Air dehumidifier	x 0: Off 1: On	0	1		
x	x	x	35860	R/W	5602	Air dehumidifier r.h. on		0	100	%	1
x	x	x	35861	R/W	5603	Air dehumidifier r.h. SD		2	50	%	1
x	x	x	35862	R/W	6710	Reset alarm relay	x 0: No 1: Yes	0	1		
x	x	x	35863	R/W	7120	Economy mode	x 0: Off 1: On	0	1		
x	x	x	35864	R/W	7120	Status/Comand	see description				
x	x	x	35865	R/W	7141	Emergency operation	x 0: Off 1: On	0	1		
x	x	x	0x9400..0x9	BL		Device					
x	x	x	37888	R/W	7700	Relay test	x 0: No test 1: Everything off 2: Relay output QX1 3: Relay output QX2 4: Relay output QX3 5: Output QX4/ZX4 6: Relay output QX5 7: Relay output QX6 8: Relay output QX7 9: Relay output QX8 10: Relay output QX9 11: Relay output QX10 12: Relay output QX11 13: Relay output QX12 14: Relay output QX13 15: Relay output QX31 16: Relay output QX32 17: Relay output QX33 18: Relay output QX34 19: Relay output QX35 20: Relay output QX21 module 1 21: Relay output QX22 module 1 22: Relay output QX23 module 1 23: Relay output QX21 module 2 24: Relay output QX22 module 2 25: Relay output QX23 module 2 26: Relay output QX21 module 3 27: Relay output QX22 module 3 28: Relay output QX23 module 3	0	28		
x	x	x	37889	R/W	7710	Output test UX1		0	100	%	1
x	x	x	37890	R/W	7710	Status/Comand	see description				
x	x	x	37891	R	7711	Output signal UX1		0	10 100	V %	1/100
x	x	x	37892	R	7711	Status	see description				
x	x	x	37893	R	7711	Unit	0..3: reserved 4: Voltage V 5: PWM %	0	5		
x	x	x	37894	R/W	7716	Output test UX2		0	100	%	1

x	x	x	37895	R/W	7716	Status/Comand	see description				
x	x	x	37896	R	7717	Output signal UX2		0	10 100	V %	1/100
x	x	x	37897	R	7717	Status	see description				
x	x	x	37898	R	7717	Unit	0..3: reserved 4: Voltage V 5: PWM %	0	5		
x		x	37899	R/W	7724	Output test UX3		0	100	%	1
x		x	37900	R/W	7724	Status	see description				
x		x	37901	R	7725	Output signal UX3		0	10 100	V %	1/100
x		x	37902	R	7725	Status	see description				
x		x	37903	R	7725	Unit	0..3: reserved 4: Voltage V 5: PWM %	0	5		
x	x	x	37904	R/W	7780	Output test UX21 module 1		0	100	%	1
x	x	x	37905	R/W	7780	Status/Comand	see description				
x	x	x	37906	R	7781	Output signal UX21 module 1		0	10 100	V %	1/100
x	x	x	37907	R	7781	Status	see description				
x	x	x	37908	R	7781	Unit	0..3: reserved 4: Voltage V 5: PWM %	0	5		
x	x	x	37909	R/W	7782	Output test UX22 module 1		0	100	%	1
x	x	x	37910	R/W	7782	Status/Comand	see description				
x	x	x	37911	R	7783	Output signal UX22 module 1		0	10 100	V %	1/100
x	x	x	37912	R	7783	Status	see description				
x	x	x	37913	R	7783	Unit	0..3: reserved 4: Voltage V 5: PWM %	0	5		
x	x	x	37914	R/W	7784	Output test UX21 module 2		0	100	%	1
x	x	x	37915	R/W	7784	Status/Comand	see description				
x	x	x	37916	R	7785	Output signal UX21 module 2		0	10 100	V %	1/100
x	x	x	37917	R	7785	Status	see description				
x	x	x	37918	R	7785	Unit	0..3: reserved 4: Voltage V 5: PWM %	0	5		
x	x	x	37919	R/W	7786	Output test UX22 module 2		0	100	%	1
x	x	x	37920	R/W	7786	Status/Comand	see description				
x	x	x	37921	R	7787	Output signal UX22 module 2		0	10 100	V %	1/100
x	x	x	37922	R	7787	Status	see description				
x	x	x	37923	R	7787	Unit	0..3: reserved 4: Voltage V 5: PWM %	0	5		
x	x	x	37924	R/W	7788	Output test UX21 module 3		0	100	%	1
x	x	x	37925	R/W	7788	Status/Comand	see description				
x	x	x	37926	R	7789	Output signal UX21 module 3		0	10 100	V %	1/100
x	x	x	37927	R	7789	Status	see description				
x	x	x	37928	R	7789	Unit	0..3: reserved 4: Voltage V 5: PWM %	0	5		
x	x	x	37929	R/W	7790	Output test UX22 module 3		0	100	%	1
x	x	x	37930	R/W	7790	Status/Comand	see description				
x	x	x	37931	R	7791	Output signal UX22 module 3		0	10 100	V %	1/100
x	x	x	37932	R	7791	Status	see description				
x	x	x	37933	R	7791	Unit	0..3: reserved 4: Voltage V 5: PWM %	0	5		
x	x	x	37934	R/W	7796	Output test WX21 module 1		0	100	%	1
x	x	x	37935	R/W	7796	Status/Comand	see description				
x	x	x	37936	R	7797	Pos step motor WX21 mod 1		0	65535		1
x	x	x	37937	R	7797	Status	see description				
x	x	x	37938	R/W	7798	Output test WX21 module 2		0	100	%	1
x	x	x	37939	R/W	7798	Status/Comand	see description				
x	x	x	37940	R	7799	Pos step motor WX21 mod 2		0	65535		1
x	x	x	37941	R	7799	Status	see description				
x	x	x	37942	R/W	7800	Output test WX21 module 3		0	100	%	1
x	x	x	37943	R/W	7800	Status/Comand	see description				
x	x	x	37944	R	7801	Pos step motor WX21 mod 3		0	65535		1
x	x	x	37945	R	7801	Status	see description				
x		x	37946	R/W	7965	Output test WX31		0	100	%	1
x		x	37947	R/W	7965	Status/Comand	see description				
x		x	37948	R	7966	Pos stepper motor WX31		0	65535		1
x		x	37949	R	7966	Status	see description				
x		x	37950	R/W	7705	Mod setp ZX6 relay test		0	100	%	1
x		x	37951	R	7708	Modulation signal ZX6		0	100	%	1
x		x	37952	R	7708	Status	see description				
x		x	37953	R/W	7969	Mod setp ZX34 relay test		0	100	%	1
x		x	37954	R	7970	Modulation signal ZX34		0	100	%	1
x		x	37955	R	7970	Status	see description				
x	x		37956	R/W	7705	Mod setp ZX4 relay test		0	100	%	1
x	x		37957	R	7708	Modulation signal ZX4		0	100	%	1

x	x		37958	R	7708	Status	see description				
x	x	x				Device Register reserve					
x	x	x	38000	R	7804	Sensor temp BX1		-28	350	°C	1/64
x	x	x	38001	R	7804	Status	see description				
x	x	x	38002	R	7805	Sensor temp BX2		-28	350	°C	1/64
x	x	x	38003	R	7805	Status	see description				
x	x	x	38004	R	7806	Sensor temp BX3		-28	350	°C	1/64
x	x	x	38005	R	7806	Status	see description				
x	x	x	38006	R	7807	Sensor temp BX4		-28	350	°C	1/64
x	x	x	38007	R	7807	Status	see description				
x		x	38008	R	7808	Sensor temp BX5		-28	350	°C	1/64
x		x	38009	R	7808	Status	see description				
x		x	38010	R	7809	Sensor temp BX6		-28	350	°C	1/64
x		x	38011	R	7809	Status	see description				
x	x		38012	R	7810	Sensor temp BX7		-28	350	°C	1/64
x	x		38013	R	7810	Status	see description				
x	x		38014	R	7811	Sensor temp BX8		-28	350	°C	1/64
x	x		38015	R	7811	Status	see description				
x	x		38016	R	7812	Sensor temp BX9		-28	350	°C	1/64
x	x		38017	R	7812	Status	see description				
x	x		38018	R	7813	Sensor temp BX10		-28	350	°C	1/64
x	x		38019	R	7813	Status	see description				
x	x		38020	R	7814	Sensor temp BX11		-28	350	°C	1/64
x	x		38021	R	7814	Status	see description				
x	x		38022	R	7815	Sensor temp BX12		-28	350	°C	1/64
x	x		38023	R	7815	Status	see description				
x	x		38024	R	7816	Sensor temp BX13		-28	350	°C	1/64
x	x		38025	R	7816	Status	see description				
x	x		38026	R	7817	Sensor temp BX14		-28	350	°C	1/64
x	x		38027	R	7817	Status	see description				
x	x	x	38028	R	7830	Sensor temp BX21 module 1		-28	350	°C	1/64
x	x	x	38029	R	7830	Status	see description				
x	x	x	38030	R	7831	Sensor temp BX22 module 1		-28	350	°C	1/64
x	x	x	38031	R	7831	Status	see description				
x	x	x	38032	R	7832	Sensor temp BX21 module 2		-28	350	°C	1/64
x	x	x	38033	R	7832	Status	see description				
x	x	x	38034	R	7833	Sensor temp BX22 module 2		-28	350	°C	1/64
x	x	x	38035	R	7833	Status	see description				
x	x	x	38036	R	7834	Sensor temp BX21 module 3		-28	350	°C	1/64
x	x	x	38037	R	7834	Status	see description				
x	x	x	38038	R	7835	Sensor temp BX22 module 3		-28	350	°C	1/64
x	x	x	38039	R	7835	Status	see description				
x		x	38040	R	7973	Sensor temp BX31		-28	350	°C	1/64
x		x	38041	R	7973	Status	see description				
x		x	38042	R	7974	Sensor temp BX32		-28	350	°C	1/64
x		x	38043	R	7974	Status	see description				
x		x	38044	R	7975	Sensor temp BX33		-28	350	°C	1/64
x		x	38045	R	7975	Status	see description				
x		x	38046	R	7976	Sensor temp BX34		-28	350	°C	1/64
x		x	38047	R	7976	Status	see description				
x		x	38048	R	7977	Sensor temp BX35		-28	350	°C	1/64
x		x	38049	R	7977	Status	see description				
x		x	38050	R	7978	Sensor temp BX36		-28	350	°C	1/64
x		x	38051	R	7978	Status	see description				
x	x	x				Device Register reserve					
x	x	x	38100	R	7844	Input signal H1		0	2^32		
			38101			(32 Bit Data)					
x	x	x	38102	R	7844	Status	see description				
x	x	x	38103	R	7844	Unit	x 0: None 1: Closed (ooo), Open (---) 2: Pulse 3: Frequency Hz 4: Voltage V	0	4		
x	x	x	38104	R	7845	Input signal H2 module 1		0	2^32		
			38105			(32 Bit Data)					
x	x	x	38106	R	7845	Status	see description				
x	x	x	38107	R	7845	Unit	x 0: None ... 4: Voltage V	0	4		
x	x	x	38108	R	7845	Input signal H21 module 1		0	2^32		
			38109			(32 Bit Data)					
x	x	x	38110	R	7845	Status	see description				
x	x	x	38111	R	7845	Unit	x 0: None ... 4: Voltage V	0	4		
x	x	x	38112	R	7846	Input signal H22 module 1		0	2^32		
			38113			(32 Bit Data)					
x	x	x	38114	R	7846	Status	see description				
			38115	R	7846	Unit	x 0: None ... 4: Voltage V	0	4		
x	x	x	38116	R	7847	Input signal H2 module 2		0	2^32		
			38117			(32 Bit Data)					
x	x	x	38118	R	7847	Status	see description				
x	x	x	38119	R	7847	Unit	x 0: None ... 4: Voltage V	0	4		

x	x	x	38120	R	7847	Input signal H21 module 2		0	2^32		
			38121				(32 Bit Data)				
x	x	x	38122	R	7847	Status	see description				
x	x	x	38123	R	7847	Unit	x 0: None ... 4: Voltage V	0	4		
x	x	x	38124	R	7848	Input signal H22 module 2		0	2^32		
			38125				(32 Bit Data)				
x	x	x	38126	R	7848	Status	see description				
x	x	x	38127	R	7848	Unit	x 0: None ... 4: Voltage V	0	4		
x	x	x	38128	R	7849	Input signal H2 module 3		0	2^32		
			38129				(32 Bit Data)				
x	x	x	38130	R	7849	Status	see description				
x	x	x	38131	R	7849	Unit	x 0: None ... 4: Voltage V	0	4		
x	x	x	38132	R	7849	Input signal H21 module 3		0	2^32		
			38133				(32 Bit Data)				
x	x	x	38134	R	7849	Status	see description				
x	x	x	38135	R	7849	Unit	x 0: None ... 4: Voltage V	0	4		
x	x	x	38136	R	7850	Input signal H22 module 3		0	2^32		
			38137				(32 Bit Data)				
x	x	x	38138	R	7850	Status	see description				
x	x	x	38139	R	7850	Unit	x 0: None ... 4: Voltage V	0	4		
x	x	x	38140	R	7858	Input signal H3		0	2^32		
			38141				(32 Bit Data)				
x	x	x	38142	R	7858	Status	see description				
x	x	x	38143	R	7858	Unit	x 0: None ... 4: Voltage V	0	4		
x		x	38144	R	7989	Input signal H31		0	2^32		
			38145				(32 Bit Data)				
x		x	38146	R	7859	Status	see description				
x		x	38147	R	7859	Unit	x 0: None ... 4: Voltage V	0	4		
x		x	38148	R	7994	Input signal H32		0	2^32		
			38149				(32 Bit Data)				
x		x	38150	R	7994	Status	see description				
x		x	38151	R	7994	Unit	x 0: None ... 4: Voltage V	0	4		
x		x	38152	R	7999	Input signal H33		0	2^32		
			38153				(32 Bit Data)				
x		x	38154	R	7999	Status	see description				
x		x	38155	R	7999	Unit	x 0: None ... 4: Voltage V	0	4		
x		x	38156	R	7728	Defrosting DI6	x 0: Off 1: On	0	1		
x		x	38157	R	7728	Status/Comand	see description				
x		x	38158	R	7729	Error/alarm message DI7	x 0: Off 1: On	0	255		
x		x	38159	R	7729	Status/Comand	see description				
x	x	x	38160	R	7911	Input EX1	x 0: 0V 1: 230V	0	1		
x	x	x	38161	R	7911	Status/Comand	see description				
x	x	x	38162	R	7912	Input EX2	x 0: 0V 1: 230V	0	1		
x	x	x	38163	R	7912	Status/Comand	see description				
x	x	x	38164	R	7913	Input EX3	x 0: 0V 1: 230V	0	1		
x	x	x	38165	R	7913	Status/Comand	see description				
x	x	x	38166	R	7914	Input EX4	x 0: 0V 1: 230V	0	1		
x	x	x	38167	R	7914	Status/Comand	see description				
x	x		38168	R	7915	Input EX5	x 0: 0V 1: 230V	0	1		
x	x		38169	R	7915	Status/Comand	see description				
x	x		38170	R	7916	Input EX6	x 0: 0V 1: 230V	0	1		
x	x		38171	R	7916	Status/Comand	see description				
x	x		38172	R	7917	Input EX7	x 0: 0V 1: 230V	0	1		
x	x		38173	R	7917	Status/Comand	see description				
x	x		38174	R	7919	Input EX9	x 0: 0V 1: 230V	0	1		
x	x		38175	R	7919	Status/Comand	see description				
x	x		38176	R	7945	Input EX10	x 0: 0V 1: 230V	0	1		
x	x		38177	R	7945	Status/Comand	see description				

x	x		38178	R	7946	Input EX11	x 0: 0V 1: 230V	0	1		
x	x		38179	R	7946	Status/Comand	see description				
x	x	x	38180	R	7950	Input EX21 module 1	x 0: 0V 1: 230V	0	1		
x	x	x	38181	R	7950	Status/Comand	see description				
x	x	x	38182	R	7951	Input EX21 module 2	x 0: 0V 1: 230V	0	1		
x	x	x	38183	R	7951	Status/Comand	see description				
x	x	x	38184	R	7952	Input EX21 module 3	x 0: 0V 1: 230V	0	1		
x	x	x	38185	R	7952	Status/Comand	see description				
x	x	x				Device Register reserve					
x		x	38400	R	7721	Heating mode D1	x 0: Off 1: On	0	1		
x		x	38401	R	7721	Status	see description				
x		x	38402	R	7722	Cooling mode D2	x 0: Off 1: On	0	1		
x		x	38403	R	7722	Status	see description				
x		x	38404	R	7723	Heat pump D3	x 0: Off 1: On	0	1		
x		x	38405	R	7723	Status	see description				
x	x	x	38406	R	9031	Relay output QX1	x 0: Off 1: On	0	1		
x	x	x	38407	R	9031	Status	see description				
x	x	x	38408	R	9032	Relay output QX2	x 0: Off 1: On	0	1		
x	x	x	38409	R	9032	Status	see description				
x	x	x	38410	R	9033	Relay output QX3	x 0: Off 1: On	0	1		
x	x	x	38411	R	9033	Status	see description				
x	x	x	38412	R	9034	Triac output ZX4/Relay output QX4	x 0: Off 1: On	0	1		
x	x	x	38413	R	9034	Status	see description				
x	x	x	38414	R	9035	Relay output QX5	x 0: Off 1: On	0	1		
x	x	x	38415	R	9035	Status	see description				
x	x	x	38416	R	9036	Triac output ZX6 / Relay output QX6	x 0: Off 1: On	0	1		
x	x	x	38417	R	9036	Status	see description				
x	x		38418	R	9037	Relay output QX7	x 0: Off 1: On	0	1		
x	x		38419	R	9037	Status	see description				
x	x		38420	R	9038	Relay output QX8	x 0: Off 1: On	0	1		
x	x		38421	R	9038	Status	see description				
x	x		38422	R	9039	Relay output QX9	x 0: Off 1: On	0	1		
x	x		38423	R	9039	Status	see description				
x	x		38424	R	9040	Relay output QX10	x 0: Off 1: On	0	1		
x	x		38425	R	9040	Status	see description				
x	x		38426	R	9041	Relay output QX11	x 0: Off 1: On	0	1		
x	x		38427	R	9041	Status	see description				
x	x		38428	R	9042	Relay output QX12	x 0: Off 1: On	0	1		
x	x		38429	R	9042	Status	see description				
x	x		38430	R	9043	Relay output QX13	x 0: Off 1: On	0	1		
x	x		38431	R	9043	Status	see description				
x	x	x	38432	R	9050	Relay output QX21 module 1	x 0: Off 1: On	0	1		
x	x	x	38433	R	9050	Status	see description				
x	x	x	38434	R	9051	Relay output QX22 module 1	x 0: Off 1: On	0	1		
x	x	x	38435	R	9051	Status	see description				
x	x	x	38436	R	9052	Relay output QX23 module 1	x 0: Off 1: On	0	1		
x	x	x	38437	R	9052	Status	see description				
x	x	x	38438	R	9053	Relay output QX21 module 2	x 0: Off 1: On	0	1		
x	x	x	38439	R	9053	Status	see description				
x	x	x	38440	R	9054	Relay output QX22 module 2	x 0: Off 1: On	0	1		
x	x	x	38441	R	9054	Status	see description				
x	x	x	38442	R	9055	Relay output QX23 module 2	x 0: Off 1: On	0	1		
x	x	x	38443	R	9055	Status	see description				
x	x	x	38444	R	9056	Relay output QX21 module 3	x 0: Off 1: On	0	1		
x	x	x	38445	R	9056	Status	see description				
x	x	x	38446	R	9057	Relay output QX22 module 3	x 0: Off 1: On	0	1		
x	x	x	38447	R	9057	Status	see description				
x	x	x	38448	R	9058	Relay output QX23 module 3	x 0: Off 1: On	0	1		

x	x	x	38449	R	9058	Status	see description				
x		x	38450	R	9071	Relay output QX31	x 0: Off 1: On	0	1		
x		x	38451	R	9071	Status	see description				
x		x	38452	R	9072	Relay output QX32	x 0: Off 1: On	0	1		
x		x	38453	R	9072	Status	see description				
x		x	38454	R	9073	Relay output QX33	x 0: Off 1: On	0	1		
x		x	38455	R	9073	Status	see description				
x		x	38456	R	9074	Triac output ZX34 / Relay output QX34	x 0: Off 1: On	0	1		
x		x	38457	R	9074	Status	see description				
x		x	38458	R	9075	Relay output QX35	x 0: Off 1: On	0	1		
x		x	38459	R	9075	Status	see description				
x	x	x				Device Register reserve					
x	x	x	38600	R		Partial diagram heat pump		0	255		
x	x	x	38601	R		Partial diagram solar collector		0	255		
x	x	x	38602	R		Partial diagram dhw storage		0	255		
x	x	x	38603	R		Partial diagram buffer		0	255		
x	x	x	38604	R		Partial diagram heat circuit 1		0	255		
x	x	x	38605	R		Partial diagram cooling circuit 1		0	255		
x	x	x	38606	R		Partial diagram heat circuit 2		0	255		
x	x	x	38607	R		Partial diagram cooling circuit 2		0	255		
x	x	x	38608	R		Partial diagram heat circuit 2		0	255		
x	x	x	38609	R		Partial diagram cooling circuit 3		0	255		
x	x	x	38610	R		Partial diagram converter		0	255		
x	x	x	38611	R		Partial diagram solid fuel boiler		0	255		
x	x	x	38612	R		Partial diagram swimming pool		0	255		
x	x	x	38613	R		Partial diagram hydraulic balancing		0	255		
x	x	x	38614	R		Partial diagram instantaneous heater		0	255		
x	x	x	38615	R		Partial diagram Consumer circuit 1		0	255		
x	x	x	38616	R		Partial diagram Consumer circuit 2		0	255		
x	x	x	38617	R		Partial diagram Consumer circuit 3		0	255		
x	x	x	38618	R		Cascade status	0: Inactiv 1: Active	0	255		
x	x	x	38619	R		Partial diagram suppl source		0	255		
x	x	x	0x9800..0x9F	BL		Device block 2					
x	x	x	38912	R	6220	Software version		0.0	99		1/10
x	x	x	38913	R	6221	Development index		0	65535		
x	x	x	38914	R		Modbus Object List Version		0.0	65535		1/10
x	x	x	38915	R	6222	Device hours run		0	199'999	h	1/3600
			38916				(32 Bit Data)				
x	x	x				Device Register reserve					
x	x	x	38928	R	6800	History 1	Year	100	199	Year	1
x	x	x	38929	R	6800		Month	1	12	Month	1
x	x	x	38930	R	6800		Day	1	31	Day	1
x	x	x	38931	R	6800		Hour	0	23	h	1
x	x	x	38932	R	6800		Minute	0	59	m	1
x	x	x	38933	R	6800	Status	see description				
x	x	x	38934	R	6801	Error code 1	ErrorCode	0	65535		
x	x	x	38935	R	6801	Status	see description				
x	x	x	38936	R	6802	History 2	Year	100	199	Year	1
x	x	x	38937	R	6802		Month	1	12	Month	1
x	x	x	38938	R	6802		Day	1	31	Day	1
x	x	x	38939	R	6802		Hour	0	23	h	1
x	x	x	38940	R	6802		Minute	0	59	m	1
x	x	x	38941	R	6802	Status	see description				
x	x	x	38942	R	6803	Error code 2	ErrorCode	0	65535		
x	x	x	38943	R	6803	Status	see description				
x	x	x	38944	R	6804	History 3	Year	100	199	Year	1
x	x	x	38945	R	6804		Month	1	12	Month	1
x	x	x	38946	R	6804		Day	1	31	Day	1
x	x	x	38947	R	6804		Hour	0	23	h	1
x	x	x	38948	R	6804		Minute	0	59	m	1
x	x	x	38949	R	6804	Status	see description				
x	x	x	38950	R	6805	Error code 3	ErrorCode	0	65535		
x	x	x	38951	R	6805	Status	see description				
x	x	x	38952	R	6806	History 4	Year	100	199	Year	1
x	x	x	38953	R	6806		Month	1	12	Month	1
x	x	x	38954	R	6806		Day	1	31	Day	1
x	x	x	38955	R	6806		Hour	0	23	h	1
x	x	x	38956	R	6806		Minute	0	59	m	1
x	x	x	38957	R	6806	Status	see description				
x	x	x	38958	R	6807	Error code 4	ErrorCode	0	65535		
x	x	x	38959	R	6807	Status	see description				
x	x	x	38960	R	6808	History 5	Year	100	199	Year	1
x	x	x	38961	R	6808		Month	1	12	Month	1
x	x	x	38962	R	6808		Day	1	31	Day	1
x	x	x	38963	R	6808		Hour	0	23	h	1
x	x	x	38964	R	6808		Minute	0	59	m	1
x	x	x	38965	R	6808	Status	see description				
x	x	x	38966	R	6809	Error code 5	ErrorCode	0	65535		
x	x	x	38967	R	6809	Status	see description				
x	x	x	38968	R	6810	History 6	Year	100	199	Year	1
x	x	x	38969	R	6810		Month	1	12	Month	1

x	x	x	38970	R	6810		Day	1	31	Day	1
x	x	x	38971	R	6810		Hour	0	23	h	1
x	x	x	38972	R	6810		Minute	0	59	m	1
x	x	x	38973	R	6810	Status	see description				
x	x	x	38974	R	6811	Error code 6	ErrorCode	0	65535		
x	x	x	38975	R	6811	Status	see description				
x	x	x	38976	R	6812	History 7	Year	100	199	Year	1
x	x	x	38977	R	6812		Month	1	12	Month	1
x	x	x	38978	R	6812		Day	1	31	Day	1
x	x	x	38979	R	6812		Hour	0	23	h	1
x	x	x	38980	R	6812		Minute	0	59	m	1
x	x	x	38981	R	6812	Status	see description				
x	x	x	38982	R	6813	Error code 7	ErrorCode	0	65535		
x	x	x	38983	R	6813	Status	see description				
x	x	x	38984	R	6814	History 8	Year	100	199	Year	1
x	x	x	38985	R	6814		Month	1	12	Month	1
x	x	x	38986	R	6814		Day	1	31	Day	1
x	x	x	38987	R	6814		Hour	0	23	h	1
x	x	x	38988	R	6814		Minute	0	59	m	1
x	x	x	38989	R	6814	Status	see description				
x	x	x	38990	R	6815	Error code 8	ErrorCode	0	65535		
x	x	x	38991	R	6815	Status	see description				
x	x	x	38992	R	6816	History 9	Year	100	199	Year	1
x	x	x	38993	R	6816		Month	1	12	Month	1
x	x	x	38994	R	6816		Day	1	31	Day	1
x	x	x	38995	R	6816		Hour	0	23	h	1
x	x	x	38996	R	6816		Minute	0	59	m	1
x	x	x	38997	R	6816	Status	see description				
x	x	x	38998	R	6817	Error code 9	ErrorCode	0	65535		
x	x	x	38999	R	6817	Status	see description				
x	x	x	39000	R	6818	History 10	Year	100	199	Year	1
x	x	x	39001	R	6818		Month	1	12	Month	1
x	x	x	39002	R	6818		Day	1	31	Day	1
x	x	x	39003	R	6818		Hour	0	23	h	1
x	x	x	39004	R	6818		Minute	0	59	m	1
x	x	x	39005	R	6818	Status	see description				
x	x	x	39006	R	6819	Error code 10	ErrorCode	0	65535		
x	x	x	39007	R	6819	Status	see description				
x	x	x	39008	R/W	6820	Reset history	x 0: No 1: Yes	0	1		
x	x	x				Device Register reserve					
x	x	x	39040	R		Error Message	ErrorCode	0	65535		
x	x	x	39041	R		Error Prio	priority	0	65535		
x	x	x	39042	R		Service Message	ServiceCode	0	65535		
x	x	x	39043	R		Error Prio	priority	0	65535		
x	x	x	39044	R		Info Sonderbetrieb	Bit7: Special operation active Bit0..6: Special operation: 0= Normal (no special operation) 1= Manual operation 2= SLT test 3= Chimney sweep function 4= Controller stop 5= Combustion optimization low-fire 6= Combustion optimization high-fire 7= Emergency operation 8= Output test 9= Simulation outside temp 10= Alternative energy mode 11= Commissioning function 12= Deareation function 13= Dry fire function 14= Eco operation 15= Initializing 16= DHW Push	0	255		
x	x	x	x9BF0..0x9FF	BL		Device block 3					
x	x	x	39920	R/W		Clock time and Date	Year	100	199	Jahr	1
x	x	x	39921	R/W		Clock time and Date	Month	1	12	Month	1
x	x	x	39922	R/W		Clock time and Date	Day	1	31	Day	1
x	x	x	39923	R/W		Clock time and Date	Hour	0	23	h	1
x	x	x	39924	R/W		Clock time and Date	Minute	0	59	min	1
x	x	x	39925	R/W		Clock time and Date	Second	0	59	s	1
x	x	x		BL		Time prog heating circuit 1					
x	x	x	39940	R/W	501, 502	Monday 1st phase on Monday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39941	R/W	503, 504	Monday 2nd phase on Monday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39942	R/W	505, 506	Monday 3rd phase on Monday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39943	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				

x	x	x	39944	R/W	501, 502	Tuesday 1st phase on Tuesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39945	R/W	503, 504	Tuesday 2nd phase on Tuesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39946	R/W	505, 506	Tuesday 3rd phase on Tuesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39947	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39948	R/W	501, 502	Wednesday 1st phase on Wednesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39949	R/W	503, 504	Wednesday 2nd phase on Wednesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39950	R/W	505, 506	Wednesday 3rd phase on Wednesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39951	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39952	R/W	501, 502	Thursday 1st phase on Thursday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39953	R/W	503, 504	Thursday 2nd phase on Thursday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39954	R/W	505, 506	Thursday 3rd phase on Thursday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39955	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39956	R/W	501, 502	Friday 1st phase on Friday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39957	R/W	503, 504	Friday 2nd phase on Friday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39958	R/W	505, 506	Friday 3rd phase on Friday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39959	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39960	R/W	501, 502	Saturday 1st phase on Saturday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39961	R/W	503, 504	Saturday 2nd phase on Saturday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39962	R/W	505, 506	Saturday 3rd phase on Saturday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39963	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39964	R/W	501, 502	Sunday 1st phase on Sunday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39965	R/W	503, 504	Sunday 2nd phase on Sunday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39966	R/W	505, 506	Sunday 3rd phase on Sunday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39967	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39968	R	516	Default values	x 0: No 1: Yes	0	1		
x	x	x		BL		Time prog heating circuit 2					
x	x	x	39980	R/W	521, 522	Monday 1st phase on Monday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39981	R/W	523, 524	Monday 2nd phase on Monday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39982	R/W	525, 526	Monday 3rd phase on Monday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min

x	x	x	39983	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39984	R/W	521, 522	Tuesday 1st phase on Tuesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39985	R/W	523, 524	Tuesday 2nd phase on Tuesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39986	R/W	525, 526	Tuesday 3rd phase on Tuesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39987	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39988	R/W	521, 522	Wednesday 1st phase on Wednesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39989	R/W	523, 524	Wednesday 2nd phase on Wednesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39990	R/W	525, 526	Wednesday 3rd phase on Wednesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39991	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39992	R/W	521, 522	Thursday 1st phase on Thursday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39993	R/W	523, 524	Thursday 2nd phase on Thursday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39994	R/W	525, 526	Thursday 3rd phase on Thursday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39995	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	39996	R/W	521, 522	Friday 1st phase on Friday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39997	R/W	523, 524	Friday 2nd phase on Friday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39998	R/W	525, 526	Friday 3rd phase on Friday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	39999	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40000	R/W	521, 522	Saturday 1st phase on Saturday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40001	R/W	523, 524	Saturday 2nd phase on Saturday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40002	R/W	525, 526	Saturday 3rd phase on Saturday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40003	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40004	R/W	521, 522	Sunday 1st phase on Sunday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40005	R/W	523, 524	Sunday 2nd phase on Sunday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40006	R/W	525, 526	Sunday 3rd phase on Sunday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40007	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40008	R/W	536	Default values	x 0: No 1: Yes	0	1		

x	x	x		BL		Time program 3/HC3					
x	x	x	40040	R/W	541, 542	Monday 1st phase on Monday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40041	R/W	543, 544	Monday 2nd phase on Monday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40042	R/W	545, 546	Monday 3rd phase on Monday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40043	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40044	R/W	541, 542	Tuesday 1st phase on Tuesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40045	R/W	543, 544	Tuesday 2nd phase on Tuesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40046	R/W	545, 546	Tuesday 3rd phase on Tuesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40047	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40048	R/W	541, 542	Wednesday 1st phase on Wednesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40049	R/W	543, 544	Wednesday 2nd phase on Wednesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40050	R/W	545, 546	Wednesday 3rd phase on Wednesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40051	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40052	R/W	541, 542	Thursday 1st phase on Thursday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40053	R/W	543, 544	Thursday 2nd phase on Thursday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40054	R/W	545, 546	Thursday 3rd phase on Thursday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40055	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40056	R/W	541, 542	Friday 1st phase on Friday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40057	R/W	543, 544	Friday 2nd phase on Friday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40058	R/W	545, 546	Friday 3rd phase on Friday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40059	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40060	R/W	541, 542	Saturday 1st phase on Saturday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40061	R/W	543, 544	Saturday 2nd phase on Saturday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40062	R/W	545, 546	Saturday 3rd phase on Saturday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40063	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40064	R/W	541, 542	Sunday 1st phase on Sunday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40065	R/W	543, 544	Sunday 2nd phase on Sunday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40066	R/W	545, 546	Sunday 3rd phase on Sunday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min

x	x	x	40067	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40068	R/W	556	Default values	x 0: No 1: Yes	0	1		
x	x	x		BL		Time program 4 / DHW					
x	x	x	40080	R/W	561, 562	Monday 1st phase on Monday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40081	R/W	563, 564	Monday 2nd phase on Monday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40082	R/W	565, 566	Monday 3rd phase on Monday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40083	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40084	R/W	561, 562	Tuesday 1st phase on Tuesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40085	R/W	563, 564	Tuesday 2nd phase on Tuesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40086	R/W	565, 566	Tuesday 3rd phase on Tuesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40087	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40088	R/W	561, 562	Wednesday 1st phase on Wednesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40089	R/W	563, 564	Wednesday 2nd phase on Wednesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40090	R/W	565, 566	Wednesday 3rd phase on Wednesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40091	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40092	R/W	561, 562	Thursday 1st phase on Thursday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40093	R/W	563, 564	Thursday 2nd phase on Thursday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40094	R/W	565, 566	Thursday 3rd phase on Thursday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40095	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40096	R/W	561, 562	Friday 1st phase on Friday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40097	R/W	563, 564	Friday 2nd phase on Friday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40098	R/W	565, 566	Friday 3rd phase on Friday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40099	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40100	R/W	561, 562	Saturday 1st phase on Saturday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40101	R/W	563, 564	Saturday 2nd phase on Saturday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40102	R/W	565, 566	Saturday 3rd phase on Saturday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min

x	x	x	40103	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40104	R/W	561, 562	Sunday 1st phase on Sunday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40105	R/W	563, 564	Sunday 2nd phase on Sunday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40106	R/W	565, 566	Sunday 3rd phase on Sunday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40107	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40108	R/W	576	Default values	x 0: No 1: Yes	0	1		
x	x	x		BL		Time program 5					
x	x	x	40140	R/W	601, 602	Monday 1st phase on Monday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40141	R/W	603, 604	Monday 2nd phase on Monday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40142	R/W	605, 606	Monday 3rd phase on Monday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40143	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40144	R/W	601, 602	Tuesday 1st phase on Tuesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40145	R/W	603, 604	Tuesday 2nd phase on Tuesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40146	R/W	605, 606	Tuesday 3rd phase on Tuesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40147	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40148	R/W	601, 602	Wednesday 1st phase on Wednesday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40149	R/W	603, 604	Wednesday 2nd phase on Wednesday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40150	R/W	605, 606	Wednesday 3rd phase on Wednesday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40151	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40152	R/W	601, 602	Thursday 1st phase on Thursday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40153	R/W	603, 604	Thursday 2nd phase on Thursday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40154	R/W	605, 606	Thursday 3rd phase on Thursday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40155	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40156	R/W	601, 602	Friday 1st phase on Friday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40157	R/W	603, 604	Friday 2nd phase on Friday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40158	R/W	605, 606	Friday 3rd phase on Friday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min

x	x	x	40159	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40160	R/W	601, 602	Saturday 1st phase on Saturday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40161	R/W	603, 604	Saturday 2nd phase on Saturday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40162	R/W	605, 606	Saturday 3rd phase on Saturday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40163	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40164	R/W	601, 602	Sunday 1st phase on Sunday 1st phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40165	R/W	603, 604	Sunday 2nd phase on Sunday 2nd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40166	R/W	605, 606	Sunday 3rd phase on Sunday 3rd phase off	H-Byte: phase on L-Byte: phase off	00:00 00:00	24:00 24:00	h:m	10min
x	x	x	40167	R/W		Status / Comand Switching phase status	Bit 0:1st phase on inactive Bit 1:1st phase off inactive Bit 2:2nd phase on inactive Bit 3:2nd phase off inactive Bit 4:3rd phase on inactive Bit 5:3rd phase off inactive Bit6-15: Unused				
x	x	x	40168	R/W	616	Default values	x 0: No 1: Yes	0	1		
x	x	x		BL		Holidays heating circuit 1					
x	x	x	40400	R/W	642	Holidays Start 1	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40401	R/W	642	Status/Comand	see description				
x	x	x	40402	R/W	643	Holidays Ende 1	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40403	R/W	643	Status/Comand	see description				
x	x	x	40404	R/W	642	Holidays Start 2	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40405	R/W	642	Status/Comand	see description				
x	x	x	40406	R/W	643	Holidays Ende 2	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40407	R/W	643	Status/Comand	see description				
x	x	x	40408	R/W	642	Holidays Start 3	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40409	R/W	642	Status/Comand	see description				
x	x	x	40410	R/W	643	Holidays Ende 3	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40411	R/W	643	Status/Comand	see description				
x	x	x	40412	R/W	642	Holidays Start 4	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40413	R/W	642	Status/Comand	see description				
x	x	x	40414	R/W	643	Holidays Ende 4	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40415	R/W	643	Status/Comand	see description				
x	x	x	40416	R/W	642	Holidays Start 5	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40417	R/W	642	Status/Comand	see description				
x	x	x	40418	R/W	643	Holidays Ende 5	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40419	R/W	643	Status/Comand	see description				
x	x	x	40420	R/W	642	Holidays Start 6	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40421	R/W	642	Status/Comand	see description				
x	x	x	40422	R/W	643	Holidays Ende 6	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40423	R/W	643	Status/Comand	see description				
x	x	x	40424	R/W	642	Holidays Start 7	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40425	R/W	642	Status/Comand	see description				
x	x	x	40426	R/W	643	Holidays Ende 7	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40427	R/W	643	Status/Comand	see description				
x	x	x	40428	R/W	642	Holidays Start 8	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40429	R/W	642	Status/Comand	see description				
x	x	x	40430	R/W	643	Holidays Ende 8	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40431	R/W	643	Status/Comand	see description				
x	x	x	40432	R/W	648	Holidays Operating level	x 0: Frost protection 1: Reduced	0	1		
x	x	x		BL		Holidays heating circuit 2					

x	x	x	40440	R/W	652	Holidays Start 1	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40441	R/W	652	Status/Comand	see description				
x	x	x	40442	R/W	653	Holidays Ende 1	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40443	R/W	653	Status/Comand	see description				
x	x	x	40444	R/W	652	Holidays Start 2	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40445	R/W	652	Status/Comand	see description				
x	x	x	40446	R/W	653	Holidays Ende 2	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40447	R/W	653	Status/Comand	see description				
x	x	x	40448	R/W	652	Holidays Start 3	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40449	R/W	652	Status/Comand	see description				
x	x	x	40450	R/W	653	Holidays Ende 3	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40451	R/W	653	Status/Comand	see description				
x	x	x	40452	R/W	652	Holidays Start 4	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40453	R/W	652	Status/Comand	see description				
x	x	x	40454	R/W	653	Holidays Ende 4	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40455	R/W	653	Status/Comand	see description				
x	x	x	40456	R/W	652	Holidays Start 5	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40457	R/W	652	Status/Comand	see description				
x	x	x	40458	R/W	653	Holidays Ende 5	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40459	R/W	653	Status/Comand	see description				
x	x	x	40460	R/W	652	Holidays Start 6	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40461	R/W	652	Status/Comand	see description				
x	x	x	40462	R/W	653	Holidays Ende 6	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40463	R/W	653	Status/Comand	see description				
x	x	x	40464	R/W	652	Holidays Start 7	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40465	R/W	652	Status/Comand	see description				
x	x	x	40466	R/W	653	Holidays Ende 7	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40467	R/W	653	Status/Comand	see description				
x	x	x	40468	R/W	652	Holidays Start 8	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40469	R/W	652	Status/Comand	see description				
x	x	x	40470	R/W	653	Holidays Ende 8	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40471	R/W	653	Status/Comand	see description				
x	x	x	40472	R/W	658	Holidays Operating level	x 0: Frost protection 1: Reduced	0	1		
x	x	x		BL		Holidays heating circuit 3					
x	x	x	40480	R/W	662	Holidays Start 1	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40481	R/W	662	Status/Comand	see description				
x	x	x	40482	R/W	663	Holidays Ende 1	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40483	R/W	663	Status/Comand	see description				
x	x	x	40484	R/W	662	Holidays Start 2	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40485	R/W	662	Status/Comand	see description				
x	x	x	40486	R/W	663	Holidays Ende 2	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40487	R/W	663	Status/Comand	see description				
x	x	x	40488	R/W	662	Holidays Start 3	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40489	R/W	662	Status/Comand	see description				
x	x	x	40490	R/W	663	Holidays Ende 3	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40491	R/W	663	Status/Comand	see description				
x	x	x	40492	R/W	662	Holidays Start 4	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40493	R/W	662	Status/Comand	see description				
x	x	x	40494	R/W	663	Holidays Ende 4	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40495	R/W	663	Status/Comand	see description				
x	x	x	40496	R/W	662	Holidays Start 5	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40497	R/W	662	Status/Comand	see description				
x	x	x	40498	R/W	663	Holidays Ende 5	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40499	R/W	663	Status/Comand	see description				
x	x	x	40500	R/W	662	Holidays Start 6	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40501	R/W	662	Status/Comand	see description				
x	x	x	40502	R/W	663	Holidays Ende 6	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40503	R/W	663	Status/Comand	see description				

x	x	x	40504	R/W	662	Holidays Start 7	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40505	R/W	662	Status/Comand	see description				
x	x	x	40506	R/W	663	Holidays Ende 7	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40507	R/W	663	Status/Comand	see description				
x	x	x	40508	R/W	662	Holidays Start 8	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40509	R/W	662	Status/Comand	see description				
x	x	x	40510	R/W	663	Holidays Ende 8	H-Byte: Month L-Byte: Day	1 1	12 31	Month Day	1
x	x	x	40511	R/W	663	Status/Comand	see description				
x	x	x	40512	R/W	668	Holidays Operating level	x 0: Frost protection 1: Reduced	0	1		
x	x	x	0xA400..0xA	BL		Process Signal (Bridging Solution see special Document)					
x	x	x	41984	R		TempAnfoEMxt	Value	0	140	°C	1/64
			41985				Attribute	0	65535		
			41986				Prio	0	16		
			41987				Lrel	0	100	%	1
			41988				AttributeExt	0	65535		
			not used								
x	x	x	41990	R/W		StatEExt	Attribute	0	65535		
			41991				LeistRel	0	100	%	1
			41992				LeistAbs	0	65535	kWh	1/10
			41993				BetStd	0	65535	h	1
			41994				AttributeExt	0	65535		
			not used								
			not used								
x	x	x	41997	R/W		ProzSperrErz	H-Byte: c8Value L-Byte: c8Attribut	0 0	100 255	%	1
x	x	x	41998	R/W		ProzZwangErz	H-Byte: c8Value L-Byte: c8Attribut	0 0	100 255	%	1
x	x	x	41999	R		TempAnfoKuehlEMxt	Wert	0	140	°C	1/64
			42000				Attribute	0	65535		
			not used								
			not used								
x	x	x	42005	R/W		StatEEKuehl	Attribute	0	65535		
			not used								
			not used								
x	x	x	42016	R/W		TempAnfo	Value	0	140	°C	1/64
			42017				Attribute	0	65535		
x	x	x	42018	R		StatEM	Attribute	0	65535		
			42019				LeistRel	0	100	%	1
			42020				LeistAbs	0	65535	kWh	1/10
x	x	x	42021	R		ProzSperrEM	H-Byte: c8Value L-Byte: c8Attribut	0 0	100 255	%	1
x	x	x	42022	R		ProzZwangEM	H-Byte: c8Value L-Byte: c8Attribut	0 0	100 255	%	1
x	x	x	42023	R		StatVM	Attribute				
x	x	x	42024	R		ProzSperrVM	H-Byte: c8Value L-Byte: c8Attribut	0 0	100 255	%	1
x	x	x	42025	R		ProzZwangVM	Wert Attribut	0 0	100 255	%	1
			not used								
			not used								
x	x	x	42032	R/W		TempAnfoKuehl	Value	0	140	°C	1/64
			42033				Attribute	0	65535		
x	x	x	42034	R		StatEMKuehl	Attribute	0	65535		
x	x	x	42035	R		StatVMKuehl	Attribute	0	65535		
			not used								
			not used								
x	x	x	42048	R/W		TempErzWert	Value	0	140	°C	1/64
			42049			Status	see description				
x	x	x	42050	R		TempSchiVor	Value	0	140	°C	1/64
			42051			Status	see description				
x	x	x	42052	R		TempPuffer	Value	0	140	°C	1/64
			42053			Status	see description				
x	x	x	42054	R/W		TempAuss	Value	-50	50	°C	1/64
			42055			Status	see description				
x	x	x	42056	R/W		Switch-on command heat pump See "Swi-on command HP stage x" in the manual	0: Control 1: Off 2: Stage1 3: Stage1 and 2	0	3		1

Error Codes (2 Byte Value)

0:No error
1:
2:
3:
4:
5:
6:
7:
8:
9:
10:Outside sensor
11:Solar sensor
12:Wind sensor
13:
14:
15:
16:
17:
18:
19:
20:Boiler sensor 1
21:
22:Boiler sensor 2
23:
24:
25:Boiler sensor solid fuel
26:Common flow sensor
27:Common flow sensor 2
28:Flue gas temp sensor
29:
30:Flow sensor 1
31:Flow sensor cooling 1
32:Flow sensor 2
33:Flow sensor HP
34:Condenser sensor
35:Source inlet sensor
36:Hot-gas sensor 1
37:Hot-gas sensor 2
38:Flow sensor prim contr
39:Evaporator sensor
40:Return sensor 1
41:
42:Return sensor 2
43:Return sensor solid fuel
44:Return sensor HP
45:Source outlet sensor
46:Return sensor cascade
47:Common return sensor
48:Refrigerant sensor liquid
49:
50:DHW sensor 1
51:

52:DHW sensor 2
53:
54:DHW flow sensor
55:St tank 2 DHW sens 1
56:St tank 2 DHW sens 2
57:DHW circulation sensor
58:DHW thermostat
59:
60:Room sensor 1
61:Room unit 1
62:Room unit 1 type
63:
64:Room unit 1 bus interr
65:Room sensor 2
66:Room unit 2
67:Room unit 2 type
68:Room sensor 3
69:Room unit 2 bus interr
70:Storage tank sensor 1
71:Storage tank sensor 2
72:Storage tank sensor 3
73:Collector sensor 1
74:Collector sensor 2
75:Bypass sensor
76:Special sensor 1
77:Air pressure sensor
78:Water pressure sensor
79:Brine sensor
80:LPB no communication
81:LPB short-circuit/comm
82:LPB address collision
83:BSB short-circuit
84:BSB address collision
85:BSB Radio communication
86:PPS short-circuit
87:PPS 2 short-circuit
88:PPS no communication
89:
90:Data loss in RAM
91:Data loss in EEPROM
92:Device electronics error
93:Change battery
94:Battery memory card
95:Time of day invalid
96:Minor SW failure
97:SW or HW failure
98:Extension module 1
99:Extension module 2
100:2 clock time masters
101:Par clock time source
102:Clock without backup
103:Communication failure
104:

105:Maintenance message
106:Source temp too low
107:Hot-gas compressor 1
108:Hot-gas compressor 2
109:Boiler temp supervision
110:Lockout SLT
111:Shutdown limit thermost
112:Lockout flue gas SLT
113:Safety shutd flue gas
114:Shutd flue gas thermo
115:Shutd flue gas sensor
116:Shutd flue gas sensor
117:Water pressure too high
118:Water pressure too low
119:Shutd water press swi
120:FI temp prectrl too low
121:Flow temp HC1
122:Flow temp HC2
123:Flow temp DHW too low
124:Boiler temp too low
125:Boiler temp too high
126:DHW charg temp
127:Legionella temp
128:Loss of flame in op
129:Wrong air supply
130:Flue gas temp too high
131:Burner lockout
132:Safety shutdown
133:Safety time exceeded
134:Common fault HP
135:Brine circuit
136:Press refrig circuit HP
137:No HP
138:No control sensor HP
139:
140:LPB address not valid
141:LPB config not consist
142:No device on LPB
143:
144:
145:PPS wrong device type
146:Configuration error
147:BMU not connected
148:Incompatible LBP unit
149:Configuration FPF
150:BMU
151:BMU internal
152:Parameterization
153:Unit locked
154:Plausibility criterion
155:Reset locked
156:
157: Boiler flow thermostat

158:Condensate
159:
160:Fan speed threshold
161:Max fan speed
162:Air pressure switch
163:Modulation valve
164:Flow press switch HC
165:Priority circuit
166:Air pressure switch
167:Heating output limits
168:
169:Sitherm Pro system
170:
171:Alarm contact 1 active
172:Alarm contact 2 active
173:Alarm contact 3 active
174:Alarm contact 4 active
175>Error out FPF active
176:Water press 2 too high
177:Water press 2 too low
178:Limit thermostat HC1
179:Limit thermostat HC2
180:Chimney sweep function
181:Controller stop function
182:Sitherm Pro drift test
183:Parameter setting mode
184:Modem function
185:Floor curing function
186:Configuration input
187:Configuration output
188:
189:
190:
191:SLT has tripped
192:
193:Start prevention
194:
195:Water refill time
196:Water refill time/week
197:
198:
199:
200:Fire/smoke alarm
201:Frost alarm
202:Supply air sensor
203:Airflow alarm
204:Fan overload
205:Pump/electric fault
206:Chiller
207:Fault cooling circuit
208:Flow supervision
209:Fault heating circuit
210:Fault SYNERGYR

211:Fault solid fuel boiler
212:Internal comm failure
213:Safety / blocking chain
214:Engine supervision
215:Fan, Valve fault
216:Boiler fault
217:Sensor fault
218:Pressure supervision
219:
220:AUX 1 alarm
221:AUX 2 alarm
222:Hi-press on HP op
223:Hi-press on start HC
224:Hi-press on start DHW
225:Low-pressure
226:Compressor 1 overlaod
227:Compressor 2 overlaod
228:Flow swi heat source
229:Press swi heat source
230:Source pump overload
231:Sensor B101
232:Sensor B102
233:Sensor B103
234:Sensor B104
235:Sensor B105
236:Sensor B106
237:Sensor B107
238:Sensor B108
239:Sensor B109
240:
241:Flow sensor yield
242:Return sensor yield
243:Swimming pool sensor
244:Fault source cascade
245:
246:
247:Defrost fault
248:
249:
250:
251:
252:
253>Error cause ambiguous
254:Unknown error code
255:See other error list
256:Heat exch thermostat
257:Casing sensor
258:Casing overtemp
259:CJC sensor
260:Flow sensor 3
261:Loss of flame Eng bu
262:Loss of fl' Supp bu
263:Eng bu BCU failure

264:Supp bu BCU failure
265:BCU failure
266:fan fault
267:Fan calibration
268:Spool valve fault
269:Spool valve calib
270:Exc temp diff h' exch
271:Diff'pressure too high
272:Diff'pressure too low
273:Conf'fault pres'sensor
274:Zero flow protection
275:Zero flow aft deaer
276:Zero flow
277:Zero flow DHW
278:Max temp rise
279:Stirl manifold thermo
280:Inner iron overtemp
281:Dyn absorber tripped
282:G83/ENS/GIM
283:Altern' overc trip
284:WCS overtemp
285:Alternat SC
286:Eng head overtemp
287:Eng head undertemp
288:Regenerat overtemp
289:WCS overt + inner iron
290:WCS overt + DA
291:WCS overt + G83
292:WCS overt + Alt
293:WCS overt + SC
294:WCS + eng head overt
295:WCS + eng head undert
296:WCS overt + Reg
297:
298:False flame Eng bu
299:False flame Supp bu
300:Eng head undert SW
301:Eng head overt SW
302:Eng head t'couple diff
303:Eng head t'couple cont
304:Eng head t' coupl lim
305:Eng under current
306:Black start failure
307:Engine stall
308:Stop resistor integrity
309:Power fail detection
310:Power mon comm fail
311:EGC comm failure
312:Bl' start fail' pump en
313:Bl' start fail' PSU sw
314:Bl' start fail' batt sw
315:Bl' start fail socket en
316:Grid voltage OOR

317:Grid frequency OOR
318:GSM comm failure
319:Check configuration
320:DHW charging sensor
321:DHW outlet sensor
322:Water press 3 too high
323:Water press 3 too low
324:BX same sensors
325:BX/e'module same sens
326:BX/m'grp same sens
327:E'module same funct
328:Mix group same funct
329:E'mod/m'grp same funct
330:BX1 no function
331:BX2 no function
332:BX3 no function
333:BX4 no function
334:BX5 no function
335:BX21 no function
336:BX22 no function
337:B1 no function
338:B12 no function
339:Coll pump Q5 missing
340:Coll pump Q16 missing
341:Coll sensor B6 missing
342:Solar DHW B31missing
343:Solar integration missing
344:Solar buffer K8 missing
345:Sol swi pool K18 missing
346:Boiler pump Q10 missing
347:Solid fuel boil comp sens
348:Solid fuel boil addr err
349:Buff valve Y15 missing
350:Buffer address error
351:Prim/sys pump addr err
352:Pr'less header addr err
353:Casc sens B10 missing
354:Special sensor 2
355:3-ph curr asymmetric
356:Flow switch consumers
357:Flow temp cooling 1
358:Soft starter
359:Div valve cool Y21 miss
360:Proc rev va Y22 miss
361:Source sens B91 miss
362:Source sens B92 miss
363:Compr sens B84 miss
364:Cool system HP wrong
365:Inst heater Q34 miss
366:Room temp sensor Hx
367:Room humidity sens Hx
368:Flow temp setp readjHx
369:External

370:Thermodynamic source
371:Flow temp HC3
372:Limit thermostat HC3
373:Extension module 3
374:Sitherm Pro calculation
375:BV stepper motor
376:Drift test limit value
377:Drift test prevented
378:Repetition internal
379:Repetition extran light
380:Repetition flame
381:Repetition startup
382:Repetition speed
383:No Repetition
384:Extraneous lighth
385:Mains undervoltage
386:Fan speed tolerance
387:Air pressure tolerance
388:DHW sensor no function
389:
390:
391:Room controller 1
392:Room controller 2
393:Room controller 3
394:Eng Bu BCU Com fail
395:Supp Bu BCU Com fail
396:Eng head t'c' cont integ
397:Eng head t'c' cont rise
398:Eng head t'c' lim integ
399:Eng head t'c' lim rise
400: Flow dir heat gen
401:EGC ARL relay defect
402:EGC ext electr missing
403:EGC Dumpload failure
404:EGC Stab Res failure
405:EGC ADC SEG OOR
406:EGC Backdumpl failure
407:EGC Backdumpl failure
408:EGC Backdumpl failure
409:Wrong EGC comm data
410:EGC eng signal failure
411:EGC Backdumpl active
412:EGC Backdumpl active
413:EGC Backdumpl active
414:EGC Backdumpl active
415:Eng frequency low
416:Eng unable to start
417:Eng RPV sensor
418:Eng RPV overtemp
419:Cooling water
420:Operation overtemp
421:Temp diff h'ex eng bu
422:State BCU Eng bu incon

423:State BCU Supp bu inco
424:Rep loss flame Eng bu
425:Rep. loss of fl' Supp bu
426:Check flue gas damper
427:Config flue gas damper
428:Boiler flow Eng bu
429:Dyn water pres too high
430:Dyn water pres too low
431:Primary exch sensor
432:Function ground missing
433:Heat exch temp
434:EGC, no link to grid
435:EGC, ARL in lock state
436:EGC, supply feedb discr
437:
438:Bus conflict LPB/BSB
439:Bus module not detect
440:
441:BX31 no function
442:BX32 no function
443:BX33 no function
444:BX34 no function
445:BX35 no function
446:BX36 no function
447:BX6 no function
448:No messag receiver 1
449:No messag receiver 2
450:No messag receiver 3
451:No messag receiver 4
452:HX1 no function
453:HX3 no function
454:HX31 no function
455:HX32 no function
456:HX33 no function
457:BX7 no function
458:Instant WH sensor
459:
460:
461:Return sensor 3
462:BX8 no function
463:BX9 no function
464:BX10 no function
465:BX11 no function
466:BX12 no function
467:BX13 no function
468:BX14 no function
469:HX21 no function
470:HX22 no function
471:HX2 no function
472:Flow sensor cooling 2
473:Flow sensor cooling 3
474:Flow temp cooling 2
475:Flow temp cooling 3

476:Suction gas sensor
477:Evapor press sensor
478:HX/e'module same sens
479:No refrigerant selected
480:Suction gas sensor EVI
481:Evap press sensor EVI
482:Evapor temp sensor EVI
483:Soft starter 2
484:Div valve cool Y45 miss
485:Div valve cool Y46 miss
486:No EI receiver 1
487:No EI receiver 2
488:Condens press sensor
489:Cascade master miss
490:Cascade source miss
491:Max evaporation temp
492:K2/modulat incompatible

Status Codes (2 Byte Value)

0	---
1	SLT has tripped
2	Fault
3	Limiter has tripped
4	Manual control active
5	Chim sweep fct, high-fire
6	Chim sweep fct, low-fire
7	Chimney sweep funct active
8	Locked, manual
9	Locked, automatic
10	Locked
11	Protective start
12	Protective start, low-fire
13	Return limitation
14	Return limitation, low-fire
15	Released
16	Released, low-fire
17	Overrun active
18	In operation
19	Released
20	Min limitation
21	Min limitation, low-fire
22	Min limitation active
23	Frost prot plant active
24	Frost protection active
25	Off
26	Emergency operation
27	Locked, externally
28	Limit source temp min
29	High-press in HP mode
30	Flow switch heat source
31	Press switch heat source
32	Limit hot-gas compr1
33	Limit hot-gas compr2
34	Limit switch-off temp max
35	Compr off time min active
36	Compens surplus heat
37	Limitation time active
38	Compr run time min active
39	Compensation heat deficit
40	Limit diff condens max
41	Limit diff condens min
42	Limit diff evap max
43	Limit diff evap min
44	Compr and electric on
45	Compressors 1 and 2 on
46	Compressor 1 on
47	Compressor 2 on
48	Frost protection HP
49	Flow active
50	Released, evap ready
51	No request

52	Frost prot collector active
53	Recooling active
54	Max st tank temp reached
55	Evaporation prot active
56	Overtemp prot active
57	Max charging temp reached
58	Charging DHW
59	Charging buffer
60	Charging swimming pool
61	Min charg temp not reached
62	Temp diff insufficient
63	Radiation insufficient
64	El charg, emergency mode
65	El charg, source protection
66	Charg el imm heater
67	Forced charging active
68	Partial charging active
69	Charging active
70	Charged, max st tank temp
71	Charged, max charging temp
72	Charged, forced temp
73	Charged, required temp
74	Part charged, required temp
75	Charged
76	Cold
77	Recooling via collector
78	Recooling via heat gen/HCs
79	Discharging prot active
80	Charg time limitation active
81	Charging locked
82	Charging lock active
83	Forced, max st tank temp
84	Forced, max charging temp
85	Forced, legionella setp
86	Forced, nominal setp
87	El charging, legionella setp
88	El charging, nominal setp
89	El charging, reduced setp
90	El charging, frost prot setp
91	El imm heater released
92	Push, legionella setp
93	Push, nominal setp
94	Push active
95	Charging, legionella setp
96	Charging, nominal setp
97	Charging, reduced setp
98	Charged, legionella temp
99	Charged, nominal temp
100	Charged, reduced temp
101	Frost prot room active
102	Floor curing function active
103	Restricted, boiler protection
104	Restricted, DHW priority

105	Restricted, buffer
106	Heating mode restricted
107	Forced draw buffer
108	Forced draw DHW
109	Forced draw source
110	Forced draw
111	Opt start ctrl+boost heating
112	Optimum start control
113	Boost heating
114	Comfort heating mode
115	Optimum stop control
116	Reduced heating mode
117	Frost protection flow active
118	Summer operation
119	24-hour Eco active
120	Setback reduced
121	Setback frost protection
122	Room temp limitation
123	SLT test active
124	Charging restricted
125	Defrost active
126	Dripping
127	Active cooling mode
128	Passive cooling mode
129	Cooling down evaporator
130	Preheating for defrost
131	Electric charging defrost
132	Forced defrost active
133	Dewpoint monitor active
134	Cooling limit OT active
135	Locking time after heating
136	Flow temp setp incr hygro
137	Heating mode
138	Cooling mode off
139	Limit switch-off temp min
140	Heating off/cooling locked
141	Boiler frost prot active
142	Recooling via DHW/HCs
143	Charged, min charging temp
144	Cooling mode restricted
145	Limit swi-off temp max cool
146	Cooling mode locked
147	Hot
148	Cooling mode ready
149	Protection mode cooling
150	Cooling mode Comfort
151	Charg DHW+buffer+swi pool
152	Charging DHW+buffer
153	Charging DHW+swi pool
154	Charging buffer+swi pool
155	Heating mode source
156	Heated, max swi pool temp
157	Heated, setpoint source

158	Heated, setpoint solar
159	Heated
160	Heating mode solar off
161	Heating mode source off
162	Heating mode off
163	Assisted firing active
164	Electric charging, forced
165	Electric charging, substitute
166	In operation for HC
167	In part load op for HC
168	In operation for DHW
169	In part load op for DHW
170	In op for HC, DHW
171	In part load op for HC, DHW
172	Locked, solid fuel boiler
173	Released for HC, DHW
174	Released for DHW
175	Released for HC
176	Locked, outside temp
177	Limit flow min dewpoint
178	Limit flow min OT
179	Flow limit reached
180	3-ph current asymmetric
181	Low-pressure
182	Fan overload
183	Compressor 1 overload
184	Compressor 2 overload
185	Source pump overload
186	Flow switch consumers
187	Operation limit OT min
188	Operation limit OT max
189	Limit source temp min water
190	Limit source temp min brine
191	Limit source temp max
192	Forced defrost compressor
193	Forced defrost fan
194	Defrost with compressor
195	Defrost with fan
196	Limit source temp min cooling
197	Electric on
198	Locked, Economy mode
199	Consumption
200	Ready
201	Standby charging
202	Frost prot cooling active
203	Full charging active
204	Locked, heating mode
205	Locked, source
206	Locked, buffer
207	Comp run time min activ,cool
208	Compr 1 and 2 on, cooling
209	Compr 1 on, cooling mode
210	Compr 2 on, cooling mode

211	Lockout position
212	Start prevention
213	Shutdown
214	Safety time
215	Startup
216	Standby
217	Home run
218	Prepurge
219	Postpurge
220	Controller stop active
221	Keep hot mode on
222	Keep hot mode active
223	Frost prot instant WH
224	Ignition
225	Settling time
226	Exotic gas operation
227	Drift test active
228	Special operation
229	Setting mode
230	Indiv boiler setting active
231	Start man drift test
232	Flue gas temp, switch-off
233	Flue gas temp, output red
234	Flue gas temp too high
235	Water pressure too low
236	Party function active
237	Transfer, legionella setpoint
238	Transfer, nominal setpoint
239	Transfer, reduced setpoint
240	Transfer active
241	Residual heat usage
242	Restratification active
243	Keep hot mode released
244	Source released
245	SLT limits output
246	Mains undervoltage
247	Temp drop protection active
248	Continuous pump operation
249	Charg opt energy, nominal
250	Charg opt energy, legio
251	Charg opt energy EU, nom
252	Charg opt energy EU, legio
253	Flow too low
254	Pumping off refrig, man
255	Collective state 255
256	Pumping off refrigerant
257	Start delay defrost
258	Compressor locked
259	Locked, source temp max
260	Locked, source temp min
261	Locked, return temp max
262	Locked, return temp min
263	Locked, flow temp max

264	Locked, flow temp min
265	Locked, cond temp max
266	Locked, evap temp min
267	Locked, hot-gas temp max
268	Limitation evap temp min
269	Limitation cond temp max
270	Limitation evap temp max

Modbus General Description	(Default value bold)
Connector	3-pole Screw terminal (A+, B-, REF)
Interface type	RS485
Termination	termination resistor selectable with DIP Switch
Modbus Version (modbus.org)	Modbus Application Protocol Specification V1.1b
	Modbus over Serial Line V1.02
Modbus Mode	RTU
Modbus Role	Slave
Slave Address	1 ...247
Baud rate	Configurable 1'200, 2'400, 4'800, 9'600, 19'200 , 38'400, 57'600, 76'800, 115'200
Start bit	1
Data bit	8
Stop bit	1 /2
Parity	Even /Odd/None
Modbus function codes	0x03 Read Holding registers
	0x06 Write Single Register (not for structured data types)
	0x10 Write multiple Register
Broadcast	not supported
Data type	Always 2 Byte per Register Signed /Unsigned 16bit & 32bit (32 bit in 2 Registers) Structured Data
Coding	Most significant first
Telegramm Length	max 44 Data byte
Response Timeout	300 ms
Parameter ACS7xx	[6651] Slave address
	[6652] Baud rate
	[6653] Parity
	[6654] Stop bit
Read/Write acces by registers	only bold register can be read/written. Non bold Register must be read/written at same time with the previous (bold) register

General Comments about Date Values

Year (8Bit Value)	1900-1999 -> 000...099 2000-2099 -> 100...199
Month	1 = January ... 12 = December
Hour	0...24
Minute	0...59
Second	0...59
Day	1...31
NY (No Year)	NY (No Year) = 0 year figure valid NY (No Year) = 1 year figure blank
ND (No Date)	ND (No Date) = 0 month / day valid ND (No Date) = 1 month / day blank
NW (No Week)	NW (No Week) = 0 weekday valid NW (No Week) = 1 weekday blank
NT (No Time)	NT (No Time) = 0 time of day valid NT (No Time) = 1 time of day blank
SUTi	SUTi = 0 standard time SUTi = 1 summer time

Time prog example

Switching program from 06:00h to 22:00h (Resolution 10min)

Monday 1st phase on	06:00h -->	H-Byte: phase on	36
Monday 1st phase off	22:00h -->	L-Byte: phase off	132
	06:10h		37

Status/Command

Description

Comment

Differs on selected action (read= status or write= command)	
Command is an enumeration type and not a Bit set (in contrast to STATUS). This means that only one Command can be delivered with each write access	
Status: (Read) -> Bitwise	
Bit0:ObjectUnused	DP not used by device application (inactive; out of operation) 'ObjectUnused' = true (=1) indicates that the data point is inactive
Bit1:SensorShortCircuit	Short-circuit of sensor (only relevant together with DeviceFailure) 'SensorShortCircuit' = true (=1) indicates that the DeviceFailure is a short-circuit of the sensor. This STATUS Bit cannot be influenced via Write Command
Bit2:Overriden	Data point in override mode (Not used for Modbus) 'Overriden' = true (=1) indicates that the data point has been overridden with 'Override' Command. With 'Override'-incapable data points, this Bit is always 0.
Bit3: Reserved	
Bit4: Reserved	
Bit5: Reserved	
Bit6:DeviceFailure	Device-internal fault (e.g. A/D converter) with impact on data point 'DeviceFailure' = true (=1) indicates a fault of the device with its impact on the data point (e.g. faulty sensor, faulty A/D converter). This STATUS Bit cannot be influenced via Write Command
Bit7: Reserved	Always 0
Comand: (Write) -> Values in decimal	
0: Release (Release of an overridden data)	Not used for Modbus
1: Write (Normal write access)	Mandatory for write operation!
2: Override (Access with override right)	Not used for Modbus
3: Reserved	
4: Reserved	
5: SetOutOfService (Puts data points out of operation)	used especially to deactivate setting values and the associated application functions
6: ResetOutOfService (Puts data points into operation)	used especially to activate setting values and the associated application functions
>7: Reserved	